

Eisenstein cocycles for Drinfeld modular varieties

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Abstract

We construct (partial) modular symbols valued in higher analogues of Beilinson–Kato elements in the motivic cohomology of powers of universal Drinfeld A -modules over Drinfeld modular varieties, for A a ring of integers in a global function field with base field of odd characteristic. We prove Hecke equivariance properties for these cocycles, for the “cohomological” Hecke action on modular symbols and the “geometric” Hecke action on the modular varieties themselves. In particular, in certain dimensions, these yield analogues of the Eisenstein property for versions of the map ϖ_N figuring in conjectures of the first author; in this new setting, this relates $\mathrm{GL}_n(A)$ and the motivic cohomology of lower-rank Drinfeld modular varieties over A . In the setting $n = 2$ and rank 1, most directly analogous to the original cyclotomic context, we recover essentially all the predicted properties in the “ ϖ side” of the first author’s eponymous conjectures, including integrality at bad primes for loops at a fixed cusp, parabolicity at “missing” cusps, and Hecke actions at ramified places.

1 Introduction

In the setting of the arithmetic of cyclotomic fields, the first named author pioneered [Sha] the study of a pair of maps of the rough form

$$\Upsilon : K_2(\mathbb{Z}[\zeta_N]) \otimes \mathbb{Z}_p \rightarrow H_1(X_1(N), \mathbb{Z}_p)/I, \quad \varpi : H_1(X_1(N))/I \rightarrow K_2(\mathbb{Z}[\zeta_N]),$$

with conjectures concerning several of their properties, most notably their mutual inverseness upon extending scalars for ϖ . Here, $X_1(N)$ denotes a closed modular curve of level fixing an exact N -torsion point, I is an Eisenstein ideal, and we have algebraic (or Milnor) K -theory on the other side. As a slogan, this “relates the boundary geometry of $\mathrm{GL}_2(\mathbb{Q})$ to the arithmetic of GL_1 of abelian extensions of $\mathbb{Q}$ ”.

The map Υ is defined using the arithmetic geometry of the modular curve, while the map ϖ is most naturally framed as a *modular symbol* construction, which at the indicated level may be defined (in one normalization) via the simple formula

$$[u : v] \mapsto \{1 - \zeta_N^u, 1 - \zeta_N^v\}$$

where $[u : v]$ for $u, v \in \mathbb{Z}/N$ (such that (u, v) has order N) is the Manin symbol parameterizing the geodesic from the cusp b/d to the cusp a/c , where $ad - bc = 1$ and $c \equiv u, d \equiv v \pmod{N}$. This needs to be shown to be well-defined, have the correct integrality properties (upon restriction to loops at one cusp, corresponding to the homology of the compact curve), and factor through the Eisenstein ideal; the latter statements were part of the original conjectures formulated in loc. cit. They have since been essentially wholly resolved, as we discuss below; the inverseness with Υ on the other hand has been proven only in a weaker form by Fukaya–Kato [FuKa].

A more conceptual approach to the existence of ϖ -type modular symbols was pioneered in the first author’s joint article [ShVe], and generalized and refined in the second author’s article [Xu], which together established all the aforementioned desiderata for the map ϖ . The idea is that the modular symbols of interest arise from root of unity specializations of modular symbols valued in the motivic cohomology of $\mathbb{G}_m^2$ (or more generally, algebraic tori); in the form given in loc. cit., these have the very simple pushforward formula

$$[v_1, \dots, v_n] \mapsto \left(v_1 \ \dots \ v_n \right)_* \{1 - z_1, \dots, 1 - z_n\} \in K_n^M(k(\mathbb{G}_m^n))/J$$

where $v_1, \dots, v_n$ are independent vectors in $\mathbb{Q}^n$ furnishing a class in the rank- n *Steinberg module* $\text{St}_n(\mathbb{Q})$ over $\mathbb{Q}$, the correct source for the generalization of the theory of modular symbols beyond GL_2 , and J is a submodule which becomes irrelevant after our intended specialization.

In the first author’s jointly authored survey [FKS], it is proposed that the “boundary geometry of $\text{GL}_2(\mathbb{Q})$ and arithmetic of GL_1 of abelian extensions” setting of the original conjectures should generalize to other fields, and to general linear groups of higher rank (and their corresponding Steinberg modules). In particular, it suggests a picture for GL_n and GL_{n-1} over function fields, where on the ϖ side, we consider maps out of $\text{St}_n(F)$, and the role of $\mathbb{G}_m^2$ is replaced by the n -fold fiber product of the universal rank- $(n-1)$ Drinfeld module over the corresponding Drinfeld modular variety. (Note that when $n-1=1$, the torsion sections cut out abelian extensions of F , just as in the cyclotomic setting.) The toric theta function $1-z$ is correspondingly replaced by theta functions defined by Kondo–Yasuda [KoYa1].

In the present article, we follow the geometric and symbolic approach of [Xu] to work out a large part of the ϖ side of this formalism for arbitrary global function fields F of odd

characteristic and in arbitrary rank, with our most complete results in the case $(\mathrm{GL}_2(F), \mathrm{GL}_1(F))$ most analogous to the classical cyclotomic setting. In fact, our construction is not restricted to ranks $(n, n-1)$; we obtain a construction with strong Hecke intertwining properties for a general rank- n Steinberg module and a rank- d Drinfeld module E for any positive integers n and d , using the self-fiber product E^n . Here, by “Hecke intertwining,” we mean that in this generality one has *two* Hecke actions:

- the *cohomological* Hecke action, for the Hecke algebra of $\mathrm{GL}_n(F)$ acting on rank- n modular symbols, and
- the *geometric* Hecke action, for the Hecke algebra of $\mathrm{GL}_d(F)$ acting via correspondences on the Drinfeld modular variety (and its universal family).

For example, in the diagonal case $n = d$, we obtain a Hecke-*equivariant* modular symbol for higher function field analogues of Beilinson–Kato elements, previously used by Kondo and Yasuda [KoYa1] to construct Euler systems in these settings; in general, we obtain intertwining statements whose resulting relations cut out Eisenstein ideals corresponding to various parabolic inductions from Levi subgroups of shape $(d, n-d)$ (when $n \geq d$, at least). The $n = d$ case also furnishes motivic versions of the p -adic “zeta maps” (in the cyclotomic setting) which are instrumental in Fukaya–Kato’s proof of that Υ and ϖ are inverse (or rather a weakening).

Remark 1.0.1. Over number fields F , straightforward prospects are limited for Υ because $\mathrm{GL}_n(F)$ rarely has algebraic structure on its locally symmetric space, and even in totally real cases where it does, the relation of automorphic forms to Galois representations is complicated by a Jacquet–Langlands transfer. Prospects for ϖ are also limited because “rank- d Drinfeld modules for F ” whose moduli encode the Langlands correspondence for $\mathrm{GL}_d(F)$ morally only exist for $d = 1$ and $F = \mathbb{Q}$ (with the Drinfeld module $\mathbb{G}_m$) and for F imaginary quadratic (with a CM elliptic curve as Drinfeld module). Note that because Drinfeld module varieties are all algebraic, Υ is also less problematic in the function field setting; using this, we hope (particularly in the rank-2, rank-1 setting) to extend the methods of [FuKa] to understand the remaining aspects of the conjectures in future work, especially since we also have already the analogues of the “zeta maps”.

1.1 Main results and article outline

We now present the main results of the article in simplified form, ignoring some technical details, and outline the background material and methods necessary to prove them. Fix a

function field F with affine function ring A of odd characteristic, with residue field k of cardinality q , and consider $E \rightarrow Y$ the universal Drinfeld module of rank d of some level.

We start by giving necessary background in Section 2.1, on Drinfeld modules and their moduli, Steinberg modules and their presentations, the formalism of Hecke operators and their actions (rather, the geometric action; the cohomological action is not defined until Section 4.1), and some technical motivic background. Because A may not have trivial class group, it is necessary also to set up a formalism of lattices, which is carried through the paper: in practice, rather than the self-fiber product E^n , one has to consider a “tensor product” construction $E \otimes \Lambda$ for some projective A -lattice Λ . However, in this introductory section, we will ignore this and assume A has trivial class group for the sake of brevity.

With the preamble out of the way, our first main construction is roughly:

Theorem 1.1.1 (Theorem 3.1.2). *There is a $\mathrm{GL}_n(F)$ -equivariant modular symbol*

$$\Theta : \mathrm{St}_n(F) \rightarrow K_n^M(k(E^n))$$

given by the formula

$$[v_1, \dots, v_n] \mapsto \left(v_1 \ \dots \ v_n \right)_* \{ \theta_1, \dots, \theta_n \}$$

for independent vectors $v_1, \dots, v_n \in A^n$, where here $\theta_1, \dots, \theta_n$ are pullbacks of Kondo–Yasuda theta functions with poles along the standard coordinate hyperplanes.

In practice, we actually use a “spread out” variant of this result valued in motivic cohomology (Theorem 3.2.1), since we need to be able to specialize this cocycle at an N -torsion section $s : Y \rightarrow E^n$, fixed by some subgroup $\Gamma \leq \mathrm{GL}_n(F)$, where N is some ideal of A . The proof of this theorem may be reduced via equivariance to checking a single relation when $n = 2$, which may be deduced from a “total residue zero” statement coming from a motivic localization sequence.

The cocycle $s^*\Theta$, on the one hand, has a cohomological rank- n Hecke action for level Γ , and on the other, has a geometric rank- d Hecke action via correspondences coming from the natural such action on the base Y . For each place $\mathfrak{p}$ disjoint from the support of N , and each positive integer exponent m , there are unramified Hecke operators

$$T_k^{(n)}(\mathfrak{p}^m), 0 \leq k \leq n; \quad T_\ell^{(d)}(\mathfrak{p}^m), 0 \leq \ell \leq d;$$

which generate the anemic part of the respective Hecke algebras, and act via the cohomological and geometric actions respectively on $s^*\Theta$.

The main technical result of the paper is the following very general anemic Hecke intertwining property:

Theorem 1.1.2 (Theorem 4.2.5). *Let $m = d - n$. If $d \geq n$, for every $0 \leq \ell \leq d$,*

$$T_\ell^{(d)}(\mathfrak{p}) \cdot s^* \Theta = \sum_k q^{(n-k)(\ell-k)} \binom{m}{\ell-k}_q T_k^{(n)}(\mathfrak{p}) s^* \Theta$$

the sum running over $\max(0, \ell - m) \leq k \leq \min(n, \ell)$, for all prime ideals $\mathfrak{p}$ not dividing N . We have the symmetric statement for $n \geq d$.

As mentioned, the listed relations cut out various Eisenstein ideals either for GL_n when $n \geq d$, or for GL_d when $d \geq n$, for parabolic subgroups with Levi of shape depending on the other index.

The actual setup and statement of this theorem are significantly more technical-looking in the main article body because of the need to parameterize everything by lattices, as previously mentioned. Even defining the cohomological Hecke action requires a good deal of technical finesse; this is accomplished in Section 4.1 via a formalism of adelic coset actions on a category of lattices, which can be viewed as an adelic refinement of the notions of “ Δ -modules” and “ Δ -module systems” used in the first author’s joint articles [ShVe] and [LSSW].

The strategy behind the proof of Theorem 4.2.5 is to prove the corresponding statement for lifts of the two kinds of Hecke operators to the universal family E^n and the modular symbol Θ , before specialization. The idea is that the action of the geometric Hecke operators $T_\ell^{(d)}(\mathfrak{p})$ on a symbol image

$$M_* \{\theta_1, \dots, \theta_n\}$$

for some matrix M , agrees with the action of a very simple fiberwise correspondence whose pullback to full-level- $\mathfrak{p}$ may be described purely as a sum of translation operators by $\mathfrak{p}$ -torsion (Corollary 4.2.2 and Proposition 4.2.3). Meanwhile, the cohomological Hecke correspondence on the same modular symbols, thanks to the $\mathrm{GL}_n(F)$ -equivariance of Θ and our adelic coset formalism, can be computed to have a similar capitulation pulled back to the same cover (Proposition 4.1.3).

Roughly, the picture is that the (capitulated) $\mathfrak{p}$ -torsion of E^n is rank- nd over $A/\mathfrak{p}$, naturally the tensor product $(A/\mathfrak{p})^n \otimes (A/\mathfrak{p})^d$. The translate sum for the geometric Hecke operators is the sum over k -planes in the rank- d tensor factor, while the sum for cohomological Hecke operators is the sum over ℓ -planes in the rank- n tensor factor. Identifying these correspondences at full level $\mathfrak{p}$ then becomes a matter of combinatorial linear algebra; the resulting relations between the corresponding class functions on $E[\mathfrak{p}]^n$ are precisely a q -analogue of the combinatorial Chu–Vandermonde identities, whose coefficients yield the coefficients in Theorem 4.2.5. The identity at full-level- $\mathfrak{p}$ then implies the identity at whatever level we started with, since equality of algebraic cycles can be checked after passing to an fppf cover.

Finally, the particular “extra” nice properties which one expects from the cyclotomic case seem to have direct analogues in the setting $n = 2, d = 1$. First, the previous constructions give *a priori* modular symbols with integrality at primes away from N , since the N -torsion separates cleanly into étale components on those fibers. By computing tame symbols via a geometric argument, we can verify that in the $n = 2, d = 1$ case, a restriction of our symbol $s^*\Theta$ (analogous to passing to the closed modular curve’s homology) is also unramified at primes dividing N , an important Selmer-theoretic aspect of the classical cyclotomic setting:

Theorem 1.1.3 (Theorem 4.3.1). *The restriction of $s^*\Theta$ to the submodule of $\text{St}_2(F)$ generated by symbols $[v_1, v_2]$ with both vectors in the same Γ -orbit has trivial tame symbol at all primes $\mathfrak{p}|N$.*

Moreover, in this setting, we have the following Eisenstein property even for dual ramified Hecke operators $U_1^{(2)}(\mathfrak{p})^*$ with $\mathfrak{p}|N$:

Theorem 1.1.4 (Theorem 4.3.2). *For any prime $\mathfrak{p}$ dividing N , $s^*\Theta$ is in the kernel of the operator $U_1^{(2)}(\mathfrak{p})^* - 1$.*

The proof imitates the analogous calculation in the classical cyclotomic setting of [Xu, §3.2]. In our present setting, because of the lattice formalism, our definition of $U_{\mathfrak{p}}^*$ is nearly tautologically what one needs to make the calculation of loc. cit. work.

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1.3 AI declaration

In accordance with the Leiden Declaration, the authors disclose use of AI (primarily Claude Opus and ChatGPT Astra) for literature search, copyediting, checking correctness of proofs, and general consultation, though all primary ideas were of human origin. The authors take full responsibility for correctness.

2 Preliminaries

2.1 Drinfeld modules and families

2.1.1 Drinfeld modules over A -schemes

Let C be a smooth projective geometrically integral curve over a finite field $k = \mathbb{F}_q$ of characteristic p , and let $F = k(C)$ be its function field. Let ∞ be a closed point of C with residue field $\mathbb{F}_{q_\infty}$, and denote by $|\cdot|_\infty$ the corresponding normalized absolute value $|\cdot|_\infty: F \rightarrow q_\infty^{\mathbb{Z}} \cup \{0\}$ on F . Let $A \subset F$ be the ring of regular functions on the affine scheme $C - \{\infty\}$.

Recall that for an A -scheme S with structure morphism $c: A \rightarrow \Gamma(S, \mathcal{O}_S)$, a *Drinfeld A -module of rank d* over S is a pair (E, φ) consisting of the additive S -group scheme E associated to a line bundle L on S and a homomorphism $\varphi: A \rightarrow \text{End}_{\mathbb{F}_q, S}(E)$ to the ring of $\mathbb{F}_q$ -linear S -endomorphisms of E such that for each $a \in A - \{0\}$, the endomorphism $\varphi(a)$ induces multiplication by $c(a)$ on $\text{Lie}(E) \cong L$, and $\varphi(a)$ is finite, locally free of degree $|a|_\infty^d$. We will also denote the endomorphism $\varphi(a)$ by $[a]$.

As usual, the a -torsion of E is denoted by $E[a]$; however, even for a non-principal ideal $\mathfrak{a} \subset A$, it makes sense to define the $\mathfrak{a}$ -torsion subscheme $E[\mathfrak{a}]$ of E as the finite flat A -module subscheme $E[\mathfrak{a}] = \bigcap_{a \in \mathfrak{a}} \ker[a]$ of E over S . Given any finite adele $a \in \widehat{A} \cap \mathbb{A}_F^\times$, we set $E[a] = E[a\widehat{A} \cap A]$.

Given Drinfeld modules E_1 and E_2 over S of the same rank, a morphism of $\mathbb{G}_a$ -bundles $f: E_1 \rightarrow E_2$ is an *isogeny* if it is finite and intertwines the A -action; in this case, its kernel is a finite subgroup of the torsion of E_1 . We say that f is an *ideal isogeny* if $\ker f$ is of the form $E_1[\mathfrak{a}]$ for some nonzero ideal $\mathfrak{a} \subset A$. We say that E_1 and E_2 are *ideally isogenous* if there exists an ideal isogeny $E_1 \rightarrow E_2$.

2.1.2 Analytic uniformization

Drinfeld modules for A can be uniformized by A -lattices analogously to the complex uniformization of abelian varieties; for example, see [Pap, Chapter 5]. We briefly recall this theory here.

Let $\mathbb{C}_\infty$ be the completion of the algebraic closure of F_∞ . An A -lattice Λ of rank d is a discrete subgroup of $\mathbb{C}_\infty$ which is also a projective A -submodule of rank d . Attached to an A -lattice Λ is an exponential function

$$e_\Lambda(z) = z \prod_{\lambda \in \Lambda - \{0\}} \left(1 - \frac{z}{\lambda}\right).$$

We consider the category of finite rank A -lattices in $\mathbb{C}_\infty$, which has morphisms $\alpha: \Lambda_1 \rightarrow \Lambda_2$ given by $\alpha \in \mathbb{C}_\infty$ such that $\alpha\Lambda_1 \subseteq \Lambda_2$. We say that the morphism α is an *isogeny* if $\alpha\Lambda_1$ has finite index in Λ_2 . To an A -lattice Λ , we can associate a Drinfeld module E over $\mathbb{C}_\infty$, with underlying group $\mathbb{C}_\infty$, such that $[a](e_\Lambda(z)) = e_\Lambda(az)$ for each $a \in A$. Thus, we have the analytic uniformization short exact sequence

$$0 \rightarrow \Lambda \rightarrow \mathbb{C}_\infty \xrightarrow{e_\Lambda} E(\mathbb{C}_\infty) \rightarrow 0,$$

where $E(\mathbb{C}_\infty)$ denotes the group of $\mathbb{C}_\infty$ -points of E . This gives an equivalence of categories as follows (cf. [Pap, Theorem 5.2.8, §5.3]).

Proposition 2.1.1. *The functor that sends a finite rank lattice to its associated Drinfeld module is a rank-preserving equivalence of categories between the category of finite rank A -lattices in $\mathbb{C}_\infty$ and the category of Drinfeld A -modules over $\mathbb{C}_\infty$. In fact, given a morphism $\alpha: \Lambda_1 \rightarrow \Lambda_2$, there is a unique morphism φ between the associated Drinfeld modules E_1 and E_2 fitting into the map of exact sequences*

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \Lambda_1 & \longrightarrow & \mathbb{C}_\infty & \longrightarrow & E_1(\mathbb{C}_\infty) & \longrightarrow & 0 \\ & & \downarrow \alpha & & \downarrow \alpha & & \downarrow \varphi & & \\ 0 & \longrightarrow & \Lambda_2 & \longrightarrow & \mathbb{C}_\infty & \longrightarrow & E_2(\mathbb{C}_\infty) & \longrightarrow & 0. \end{array}$$

In particular, homothety classes of A -lattices of rank d are in bijection with isomorphism classes of Drinfeld A -modules of rank d .

In the reverse direction to Proposition 2.1.1, given a Drinfeld module E over $\mathbb{C}_\infty$, we can consider the kernel of the exponential map $\text{Lie}(E) \rightarrow E(\mathbb{C}_\infty)$, which becomes a lattice $\Lambda(E)$ in $\mathbb{C}_\infty$ upon choosing an isomorphism $\text{Lie}(E) \cong \mathbb{C}_\infty$ of $\mathbb{C}_\infty$ -vector spaces (or, equivalently, upon trivializing the underlying line bundle). Given a morphism $\varphi: E_1 \rightarrow E_2$ of Drinfeld modules over $\mathbb{C}_\infty$, we use $\varphi_*: \Lambda(E_1) \rightarrow \Lambda(E_2)$ to denote the resulting morphism of lattices.

We shall suppose for convenience that our Drinfeld modules have underlying line bundle $\mathbb{C}_\infty$. In analogy with abelian varieties, we then refer to $\Lambda(E)$ as the *homology lattice* of E .

For $a \in A$, the morphism $[a]_*$ is multiplication by a on $\Lambda(E)$. For any nonzero ideal $\mathfrak{a}$ of A , we can canonically identify $E[\mathfrak{a}]$ with $\mathfrak{a}^{-1}\Lambda(E)/\Lambda(E)$. Moreover, the $\mathbb{C}_\infty$ -points of the kernel of a nonzero morphism $\varphi: E_1 \rightarrow E_2$ may be canonically identified with $\text{coker } \varphi_* = \Lambda(E_2)/\varphi_*(\Lambda(E_1))$.

2.1.3 Drinfeld modular varieties and universal families

Fix a positive integer r . Let N be a nonzero proper ideal of A , and set $U_N = \text{Spec } A - \text{Supp } N$. A *level* is an A -submodule

$$\mathcal{L} \subset (N^{-1}A/A)^r \subset (F/A)^r$$

such that N is minimal for $\mathcal{L}$. Let $\hat{A}$ denote the profinite completion of A . We define $\hat{\Gamma}_{\mathcal{L}}$ as the stabilizer of $\mathcal{L}$ under the action of $\text{GL}_r(\hat{A})$ on $(F/A)^r$, and we let $\hat{\Gamma}_{1,\mathcal{L}}$ denote the normal subgroup of $\hat{\Gamma}_{\mathcal{L}}$ fixing each element of $\mathcal{L}$.

For the level

$$\bar{N} := (A^{r-1} \times N^{-1}A)/A^r,$$

we write $\hat{\Gamma}_0^{(r)}(N)$ for $\hat{\Gamma}_{\bar{N}}$ and $\hat{\Gamma}_1^{(r)}(N)$ for $\hat{\Gamma}_{1,\bar{N}}$. In this case, the quotient $\hat{\Gamma}_0^{(r)}(N)/\hat{\Gamma}_1^{(r)}(N)$ is canonically identified with $(A/N)^\times$ via reduction of the (r, r) -entry.

We define the *finite adelic space* for GL_r/F and level $\mathcal{L}$ as

$$S_{\mathcal{L}}^{(r),f} = \text{GL}_r(F) \backslash \text{GL}_r(\mathbb{A}_F^f) / \hat{\Gamma}_{1,\mathcal{L}}.$$

This can be interpreted as a set of isomorphism classes of lattices with level structure:

Proposition 2.1.2. *There is a bijection between $S_{\mathcal{L}}^{(r),f}$ and the set of isomorphism classes of pairs (T, ι) with T a projective A -module of rank r and*

$$\iota: \mathcal{L} \rightarrow T \otimes_A F/A$$

an A -linear injection. Here, (T, ι) gives rise to the double coset of $\gamma \in \text{GL}_r(\mathbb{A}_F^f)$ such that $T \otimes_A \hat{A} = \gamma \hat{A}^r$ and $\gamma|_{\mathcal{L}} = \iota$ for the map

$$\gamma: (F/A)^r \xrightarrow{\sim} T \otimes_A (F/A)$$

induced by γ .

For $\mathcal{L} = \bar{N}$, we simply write $S_N^{(r),f}$. Note that we have $S_1^{(1),f} = \text{Pic}(A)$, the Picard group of A . For any $r \geq 1$ and any $\mathcal{L}$, the determinant map induces a map

$$\det: S_{\mathcal{L}}^{(r),f} \rightarrow S_1^{(1),f} = \text{Pic}(A).$$

For $r \geq 2$, the strong approximation theorem for SL_r tells us that for $\mathcal{L} = \bar{N}$ this is a *bijection*, since $\det(\hat{\Gamma}_0^{(r)}(N)) = \hat{A}^\times$. In other words, adding $\bar{N}$ -level structure or raising the dimension does not change the number of isomorphism classes: there are always exactly $|\text{Pic}(A)|$ -many, the same as for rank-1 projective A -modules, and we even have canonical identifications of the isomorphism classes with those of this simplest case by taking determinants.

We now specialize to $r = d$ and the setting of Drinfeld modules. Define a *Drinfeld $\mathcal{L}$ -level structure*, or *level structure*, on a Drinfeld module E over an A -scheme S to be a morphism $\iota: \mathcal{L}_S \rightarrow E$ of A -module schemes, where $\mathcal{L}_S$ is the constant group scheme attached to $\mathcal{L}$ over S , such that the Cartier divisor given by $\sum_{x \in \mathcal{L}} [\iota(x)]$ is a closed subgroup scheme of $E[N]$.

For instance, given an A -lattice Λ of rank d in $\mathbb{C}_\infty$, an $\mathcal{L}$ -level structure on the corresponding Drinfeld module is an injective map $\iota: \mathcal{L} \rightarrow \mathbb{C}_\infty/\Lambda$ of A -modules; in general, over U_N , the Drinfeld condition above reduces to ι being a monomorphism. A morphism $(E, \iota) \rightarrow (E', \iota')$ of Drinfeld modules with $\mathcal{L}$ -level structures is an isogeny $\varphi: E \rightarrow E'$ such that $\iota' = \varphi \circ \iota$. When $\mathcal{L} = \bar{N}$, we will simply refer to these as N -level structures.¹

Let $\mathcal{M}_{\mathcal{L}}^{(d)}$ denote the functor that associates to an A -scheme S the set of isomorphism classes of pairs (E, ι) with E a Drinfeld module over S and ι a Drinfeld $\mathcal{L}$ -level structure on E . We have the following extension of a theorem of Drinfeld (see [KoYa1]):

Theorem 2.1.3. *Let $\mathcal{L}$ be a nonzero level. The functor $\mathcal{M}_{\mathcal{L}}^{(d)}$ is representable over $U_{\mathfrak{f}}$ by a smooth affine scheme, where $\mathfrak{f} = \text{Ann}_A(\mathcal{L})$ is the conductor of $\mathcal{L}$. If the support of this conductor contains at least two primes, then $\mathcal{M}_{\mathcal{L}}^{(d)}$ is representable over $\text{Spec } A$ by a finite-dimensional affine scheme, smooth away from $\mathfrak{f}$. In either case, we will denote the corresponding scheme by $Y_{\mathcal{L}}^{(d)}$.*

We refer to $Y_{\mathcal{L}}^{(d)}$ as a Drinfeld modular scheme. As in the setting of arithmetic moduli of elliptic curves, we have a *universal Drinfeld module* $\mathcal{E}_{\mathcal{L}}^{(d)}$ over $Y_{\mathcal{L}}^{(d)}$, along with its tautological level structure $\iota_{\mathcal{L}}^{(d)}$. We denote $Y_{\bar{N}}^{(d)}$ more simply by $Y_N^{(d)}$.

We now pass to the analytic uniformization of $Y_{\mathcal{L}}^{(d)}$: let $\tilde{\Omega}_d$ be the complement of the union of F_∞ -rational hyperplanes through the origin in $\mathbb{C}_\infty^d$, which may be viewed as the space of injective F_∞ -linear maps from F_∞^d to $\mathbb{C}_\infty$. The *Drinfeld period domain* of dimension $d - 1$ is

$$\Omega_d = \tilde{\Omega}_d / \mathbb{C}_\infty^\times.$$

Note that Ω_d is a point for $d = 1$ and the complement of the union of the F_∞ -rational hyperplanes inside $\mathbb{P}^{d-1}(\mathbb{C}_\infty)$ for $d \geq 2$.

We set $S^{(d)}(\mathbb{C}_\infty) = \text{GL}_d(F) \backslash \text{GL}_d(\mathbb{A}_F^f) \times \Omega_d$ and define the rigid locally symmetric space for $\text{GL}_{d/F}$ of level $\mathcal{L}$ by

$$S_{\mathcal{L}}^{(d)}(\mathbb{C}_\infty) = S^{(d)}(\mathbb{C}_\infty) / \hat{\Gamma}_{1, \mathcal{L}}.$$

¹The reader may notice that these are really “ $\Gamma_1(N)$ -level structures” in the terminological convention coming from the classical setting of modular curves, but we adopt this shorter terminology since it is the sort of level structure we will consider for the most part in this article.

Strong approximation tells us that there is a bijection between the connected components of $S_{\mathcal{L}}^{(d)}(\mathbb{C}_{\infty})$ as a rigid analytic space and $S_{\mathcal{L}}^{(d),f}$. We have

$$S_{\mathcal{L}}^{(d)}(\mathbb{C}_{\infty}) = \coprod_{\gamma} \Gamma_{\mathcal{L}}^{\gamma} \backslash \Omega_d,$$

where γ runs over a set of representatives of $S_{\mathcal{L}}^{(d),f}$, and $\Gamma_{\mathcal{L}}^{\gamma} = \mathrm{GL}_d(F) \cap \gamma \hat{\Gamma}_{1,\mathcal{L}} \gamma^{-1}$.

For an A -lattice T of rank d , a choice of ordered basis $(t_1, \dots, t_d)$ of $T \otimes_A F$ determines an isomorphism of $\mathrm{Aut}_A(T \otimes_A F)$ with $\mathrm{GL}_d(F)$. So, given such a choice, we may identify the A -linear automorphisms $\mathrm{Aut}_A(T)$ of T with a subgroup of $\mathrm{GL}_d(F)$. The choice of basis affects $\mathrm{Aut}_A(T)$ in this identification only up to conjugacy. This choice also determines a $\mathrm{GL}_d(F_{\infty})$ -equivariant bijection from Ω_d to the homothety classes of A -linear injections $T \hookrightarrow \mathbb{C}_{\infty}$ with discrete image, taking $(z_i)_{i=1}^d \in \Omega_d$ to the restriction of the F -linear map taking t_i to z_i for $1 \leq i \leq d$. From this, one can see the following.

Proposition 2.1.4. *Let (T, ι) correspond by Proposition 2.1.2 to the class of $\gamma \in \mathrm{GL}_d(\mathbb{A}_F^f)$ in $S_{\mathcal{L}}^{(d),f}$. A choice of ordered basis $(t_i)_{i=1}^d$ of $T \otimes_A F$ determines a bijection between $\Gamma_{\mathcal{L}}^{\gamma} \backslash \Omega_d$ and the similarity classes of pairs (Λ, j) of A -lattices Λ of rank d in $\mathbb{C}_{\infty}$ and injections $j: \mathcal{L} \rightarrow \mathbb{C}_{\infty}/\Lambda$ that are isomorphic to (T, ι) in the sense that there exists an A -linear injection of T into $\mathbb{C}_{\infty}$ with image Λ such that ι composed with the induced map $T \otimes_A F/A \rightarrow \mathbb{C}_{\infty}/\Lambda$ equals j .*

Combining Proposition 2.1.4 with Proposition 2.1.1 gives the following.

Corollary 2.1.5. *There is a bijection $Y_{\mathcal{L}}^{(d)}(\mathbb{C}_{\infty}) \xrightarrow{\sim} S_{\mathcal{L}}^{(d)}(\mathbb{C}_{\infty})$.*

For $d \geq 2$, the algebraic closure of F in the function field of $Y_{\mathcal{L}}^{(d)}$ is the finite abelian extension of F corresponding to $F^{\times} \backslash \mathbb{A}_F^{\times} / (F_{\infty}^{\times} \det \hat{\Gamma}_{1,\mathcal{L}})$. This is also the smallest extension of F over which all of the geometric components of $Y_{\mathcal{L}}^{(d)}$ are individually defined (cf. [BBP]). For $\mathcal{L} = \bar{N}$, these components are indexed by $\mathrm{Pic}(A)$, and the corresponding extension is the maximal abelian extension of F in which ∞ splits completely.

2.1.4 Theta functions

Write $\mathbb{Z}' := \mathbb{Z}[(q_{\infty}^d - 1)^{-1}]$; this will frequently be our coefficient ring in this article. Let E be a Drinfeld module over a reduced A -scheme S . We view S as a subscheme of E via the zero section. In [KoYa1, §2.2], Kondo and Yasuda exhibited a theta function $\theta \in \mathcal{O}(E - S)^{\times} \otimes \mathbb{Z}'$, uniquely characterized by the properties:

- i. θ has a simple zero along the zero section, and

ii. $[a]_*\theta = \theta$ for all $a \in A \setminus \{0\}$,

where $[a]_*$ is the norm map for a induced by the composition

$$\mathcal{O}(E - S)^\times \hookrightarrow \mathcal{O}(E - a^{-1}S)^\times \rightarrow \mathcal{O}(E - S)^\times,$$

the first map being restriction and the second pushforward by multiplication by a .

Lemma 2.1.6. *Let $\bar{s}$ be a geometric point of S . For any $a \in A - \{0\}$ such that $c(a)$ is invertible on S , we have*

$$\prod_{t \in E_{\bar{s}}[a] - \{0\}} \theta(t) = c(a)(\bar{s}). \quad (2.1)$$

Proof. We work on the geometric fiber $E_{\bar{s}}$ of the residue field κ of $\bar{s}$, and let T be a coordinate function on the affine line $E_{\bar{s}}$. Let $P(T) \in \kappa[[T]]$ be the power series defining $[a]^*(T)$, which has T -coefficient $c(a)(\bar{s})$. Pulling back $[a]_*\theta = \theta$ by a , we have $\theta(P(T)) = \prod_{t \in E_{\bar{s}}[a]} \theta(T + t)$. Since θ has a zero of order 1 at 0, we have $\theta(T) = \lambda T$ for some $\lambda \in \kappa[T, T^{-1}]^\times \otimes \mathbb{Z}'$. This implies that $\theta(T)^{-1}\theta(P(T)) = T^{-1}P(T)$. Therefore, setting $T = 0$, we have

$$\prod_{t \in E_{\bar{s}}[a] - \{0\}} \theta(t) = \frac{P(T)}{T} \Big|_{T=0} = P'(0) = c(a)(\bar{s}).$$

□

For any nonzero $a \in A$ with $|a|_\infty^d > 1$, we also have from [KoYa1, (2.2)] the explicit formula

$$\theta|_{|a|_\infty^d - 1} = \frac{T|_{|a|_\infty^d}}{[a]_*T} \quad (2.2)$$

Zariski locally on S , for T the local coordinate of E over this neighborhood.

The theta function θ is the analogue of a theta function of an elliptic curve. A theta function over a universal Drinfeld module $\mathcal{E}$ can be pulled back by a torsion section to obtain *Kato-Siegel units* on Drinfeld modular schemes, much as Siegel units on modular curves are defined as pullbacks from universal elliptic curves. In [KoYa1], Kondo and Yasuda use d -fold products of these units to construct “zeta elements” on S , analogues of Beilinson-Kato elements for modular curves, related to L -values of modular forms. In our geometric setting of n -fold Drinfeld modules, we will construct analogous cup products of n such units; the cases $n = d$ (recovering Kondo–Yasuda) and $n = d + 1$ will be of special interest.

2.1.5 Adelic formalism for Hecke operators

We set up the general adelic formalism for Hecke operators, which will be used both for *geometric* Hecke operators on our spaces $Y_N^{(d)}$ and *cohomological* operators corresponding to dimension n .

Let r be a positive integer. We have an adelic monoid

$$\hat{\Delta}^{(r)} = M_r(\hat{A}) \cap \mathrm{GL}_r(\mathbb{A}_F^f).$$

To incorporate level- $\overline{N}$ structure, we define $\hat{\Delta}_N^{(r)} \subset \hat{\Delta}^{(r)}$ to be the submonoid of elements that fix $\hat{A}^r$ and $\hat{A}^{r-1} \times N^{-1}\hat{A}$ and induce the identity map on $\overline{N}$. Let $E_N^{(r)}$ be a set of representatives of the double cosets $\hat{\Gamma}_1^{(r)}(N)g\hat{\Gamma}_1^{(r)}(N)$ with $g \in \hat{\Delta}_N^{(r)}$.

The abstract Hecke algebra of level N is the ring with underlying abelian group freely generated by the double cosets $\hat{\Gamma}_1^{(r)}(N)g\hat{\Gamma}_1^{(r)}(N)$ for $g \in E_N^{(r)}$, on which the multiplication is defined on $g, h \in \mathrm{GL}_r(\mathbb{A}_F^f)$ by

$$[\hat{\Gamma}_1^{(r)}(N)g\hat{\Gamma}_1^{(r)}(N)] \cdot [\hat{\Gamma}_1^{(r)}(N)h\hat{\Gamma}_1^{(r)}(N)] = \sum_{t \in E_N^{(r)}} c_{g,h,t} [\hat{\Gamma}_1^{(r)}(N)t\hat{\Gamma}_1^{(r)}(N)],$$

where if L_g denotes a set of representatives of the left $\hat{\Gamma}_1^{(r)}(N)$ -cosets in $\hat{\Gamma}_1^{(r)}(N)g\hat{\Gamma}_1^{(r)}(N)$, then $c_{g,h,t}$ is the number of pairs $(x, y) \in L_g \times L_h$ such that $xy \in \hat{\Gamma}_1^{(r)}(N)t$, for the fixed left coset $\hat{\Gamma}_1^{(r)}(N)t$ inside the indicated double coset.

For a prime $\mathfrak{p}$ in A , let $\pi_{\mathfrak{p}}$ denote an idele that is a local uniformizer in the $\mathfrak{p}$ -coordinate and 1 in the others. For $\mathfrak{p} \nmid N$ and $0 \leq k \leq r$, we set

$$T_k^{(r)}(\mathfrak{p}^m) = \left[\hat{\Gamma}_1^{(r)}(N) \begin{pmatrix} \pi^m I_k & \\ & I_{r-k} \end{pmatrix} \hat{\Gamma}_1^{(r)}(N) \right].$$

For $\mathfrak{p} \mid N$ and $0 \leq k \leq r-1$, we define $U_k^{(r)}(\mathfrak{p}^m)$ by the above formula for $T_k^{(r)}(\mathfrak{p}^m)$. We can then define a total Hecke operator for $m \geq 0$ by

$$T^{(r)}(\mathfrak{p}^m) = \sum_{\substack{i_1 \geq \dots \geq i_r \geq 0 \\ i_1 + \dots + i_r = m}} \left[\hat{\Gamma}_1^{(r)}(N) \mathrm{diag}(\pi^{i_1}, \pi^{i_2}, \dots, \pi^{i_r}) \hat{\Gamma}_1^{(r)}(N) \right]$$

for $\mathfrak{p} \nmid N$ and define $U^{(r)}(\mathfrak{p}^m)$ by the modification of the latter formula to require $i_r = 0$. The definitions of these Hecke operators are independent of the choice of π .

The elements $T_k^{(r)}(\mathfrak{p})$ with $1 \leq k \leq r$ for a given $\mathfrak{p} \nmid N$ are algebraically independent. Let us denote the *anemic Hecke algebra* generated by these operators for primes $\mathfrak{p} \nmid N$ by $\mathbb{T}_N^{(r)}$. For

a prime $\mathfrak{p} \nmid N$, the $T_k^{(r)}(\mathfrak{p}^m)$ with $m \geq 1$ are in the Hecke algebra generated by these double cosets: in fact, in $\mathbb{T}_N^{(r)}[[s]]$, we have

$$\sum_{i=0}^{\infty} T_k^{(r)}(\mathfrak{p}^i) s^i = \left(\sum_{k=0}^r (-1)^k [A : \mathfrak{p}]^{k(k-1)/2} T_k^{(r)}(\mathfrak{p}) s^k \right)^{-1}.$$

2.1.6 Geometric Hecke operators

We next turn to Hecke operators as arising from correspondences on the Drinfeld modular variety $Y_N^{(d)}$ and the universal Drinfeld module $\mathcal{E}_N^{(d)}$ over it. First, we have the *diamond operators* $\langle a \rangle$ for $a \in (A/N)^\times$, which on the level of moduli send $(E, \iota) \mapsto (E, \iota \circ [a])$ where $[a]: \bar{N} \rightarrow \bar{N}$ denotes the multiplication action.

Let $U = U_N$ if N is a prime power and $U = \operatorname{Spec} A$ otherwise. For $\mathfrak{p} \nmid N$, consider the moduli functor $\mathcal{M}_{N,k,\mathfrak{p}}^{(d)}$ that on a U -scheme S is the set of isomorphism classes of triples (E, ι, C) with E a Drinfeld module of rank d over S , ι an N -structure on E , and C an S -group subscheme of E which admits k linearly independent $\mathfrak{p}$ -torsion generators as a Cartier divisor. If we invert $\mathfrak{p}$ on the base, the latter condition of being a *Drinfeld basis* becomes equivalent to asking that C be finite étale and locally isomorphic to $(A/\mathfrak{p})_S^k$. The functor $\mathcal{M}_{N,k,\mathfrak{p}}^{(d)}$ is representable by a moduli scheme $Y_{N,k,\mathfrak{p}}^{(d)}$, and there is a universal Drinfeld module $\mathcal{E}_{N,k,\mathfrak{p}}^{(d)}$ over it.

We define $\phi = \phi_k(\mathfrak{p})$ and $\psi = \psi_k(\mathfrak{p}): Y_{N,k,\mathfrak{p}}^{(d)} \rightarrow Y_N^{(d)}$ by $\phi(E, \iota, C) = (E, \iota)$ and $\psi(E, \iota, C) = (E/C, \iota')$, where ι' is the composition of ι with the quotient map $E \rightarrow E/C$.

We also have morphisms $\Phi = \Phi_k(\mathfrak{p})$ and $\Psi = \Psi_k(\mathfrak{p}): \mathcal{E}_{N,k,\mathfrak{p}}^{(d)} \rightarrow \mathcal{E}_N^{(d)}$, where Φ lifts ϕ and is the identity on fibers, and Ψ lifts ψ and is given by the map $E \rightarrow E/C$ on fibers. In particular, these fit into a commutative diagram

$$\begin{array}{ccccc} \mathcal{E}_N^{(d)} & \xleftarrow{\Psi} & \mathcal{E}_{N,k,\mathfrak{p}}^{(d)} & \xrightarrow{\Phi} & \mathcal{E}_N^{(d)} \\ \downarrow & & \downarrow & & \downarrow \\ Y_N^{(d)} & \xleftarrow{\psi} & Y_{N,k,\mathfrak{p}}^{(d)} & \xrightarrow{\phi} & Y_N^{(d)}. \end{array}$$

We shall use $T_k^{(d)}(\mathfrak{p})^*$ to denote the dual correspondence $\phi_* \circ \psi^*$ on motivic cohomology groups of $Y_N^{(d)}$, as well as to denote $\Phi_* \circ \Psi^*$ on direct limits of motivic cohomology groups of open subschemes of $\mathcal{E}_N^{(d)}$ on which Ψ^* and Φ_* are well-defined.

2.1.7 n -fold Drinfeld modules

To study cup products of n theta functions, it will be useful to work on n th powers, or “almost powers,” of Drinfeld modules. Let $E \rightarrow S$ be any Drinfeld A -module, and let $\Lambda \subset F^n$ be

a full-rank A -lattice. Then we define the corresponding n -fold Drinfeld A -module to be the contravariant functor from S -schemes to A -modules given by

$$\Lambda \otimes_A E: \kappa \mapsto \Lambda \otimes_A E(\kappa)$$

for an S -scheme or morphism between two S -schemes κ .² Since Λ is a projective A -module, this is an fppf sheaf, and is in fact representable:

Example 2.1.7. If $\Lambda = \mathfrak{a}_1 \times \dots \times \mathfrak{a}_n$ for an n -tuple of (fractional) ideals of A , then the functor $\Lambda \otimes_A E$ is represented by $E_1 \times_S \dots \times_S E_n$, where $E_i = E/E[\alpha_i \mathfrak{a}_i^{-1}]$ for some nonzero $\alpha_i \in \mathfrak{a}_i$ for $1 \leq i \leq n$. Note that the choice of α_i only changes the resulting functor of points by an isomorphism given by a diagonal matrix in $\mathrm{GL}_n(F)$.

Proposition 2.1.8. *The functor $\Lambda \otimes_A E$ is represented by a scheme whose underlying space is a rank- n vector bundle over S .*

Proof. Choose an A -module isomorphism $\Lambda \cong A^{n-1} \oplus \mathfrak{a}$ for some ideal $\mathfrak{a}$ of A . Then the functor $\Lambda \otimes_A E$ is represented by $E^{n-1} \times E/E[\mathfrak{a}]$, which is a rank- n vector bundle as a product of n line bundles over S . $\square$

We will write $T_E(\Lambda)$ for the S -scheme representing $\Lambda \otimes_A E$, omitting the E if we are working in a section where E is fixed. It carries an A -module action over S of the matrix algebra $\mathrm{End}_A(\Lambda) \subset M_n(F)$ via its obvious action on the functor of points.

As the notation suggests, $T_E(-)$ can be viewed as a functor: if $\mathbf{Lat}(n)$ is the category of full A -lattices in F^n , with morphisms $\Lambda_1 \rightarrow \Lambda_2$ given by elements $g \in \mathrm{GL}_n(F)$ for which $g\Lambda_1 \subset \Lambda_2$, then we have a map

$$T_E(-): \mathbf{Lat}(n) \rightarrow \mathbf{Isog}(E^n) \tag{2.3}$$

to the isogeny category of the n -fold Drinfeld module E^n , whose action on a morphism $g: \Lambda_1 \rightarrow \Lambda_2$ is the map of functors of points $g \otimes \mathrm{id}: \Lambda_1 \otimes_A E(-) \rightarrow \Lambda_2 \otimes_A E(-)$.

We can also add “level structure” to this construction: write $\mathbf{Lat}_N(n)$ for the category whose objects are pairs $(\Lambda, \iota: N^{-1}A/A \xrightarrow{\sim} \Lambda'/\Lambda)$ for $\Lambda \subset \Lambda'$ a pair of A -lattices in F^n , and whose morphisms

$$(\Lambda_1, \iota_1) \rightarrow (\Lambda_2, \iota_2)$$

are given by elements $g \in \mathrm{GL}_n(F)$ such that $g\Lambda_1 \subset \Lambda_2$, $g\Lambda'_1 \subset \Lambda'_2$, and the postcomposition of $g: \Lambda'_1/\Lambda_1 \rightarrow \Lambda'_2/\Lambda_2$ with ι_1 is ι_2 .

²In the setting of CM abelian varieties, the analogue of this construction is sometimes referred to as “Serre’s tensor product construction.”

Similarly, write $\mathbf{Isog}_N(E^n)$ for the category whose objects are pairs $(T, q: E[N] \hookrightarrow T)$ for T an n -fold Drinfeld module in the isogeny class of E^n and q an inclusion of A -module schemes, and whose morphisms are isogenies $T \rightarrow T'$ which commute with their respective level structures. Then we have a functor

$$T_E^{(N)}(-): \mathbf{Lat}_N(n) \rightarrow \mathbf{Isog}_N(E^n) \quad (2.4)$$

such that

$$T_E^{(N)}(\Lambda, \iota: N^{-1}A/A \rightarrow \Lambda'/\Lambda) = (T_E(\Lambda), q_\iota: E[N] \hookrightarrow T_E(\Lambda)),$$

where q_ι is the composition

$$E[N] \xrightarrow{\sim} \mathrm{Tor}_1^A(N^{-1}A/A, E) \xrightarrow{\sim} \mathrm{Tor}_1^A(\Lambda'/\Lambda, E) \hookrightarrow T_E(\Lambda)$$

induced by ι through its identification of the two Tor-groups, the first and last maps being connecting maps for the sequences induced by the surjections $E \rightarrow N^{-1}A \otimes_A E$ and $T_E(\Lambda) \rightarrow T_E(\Lambda')$, respectively. On morphisms $g: (\Lambda_1, \iota_1) \rightarrow (\Lambda_2, \iota_2)$, we set $T_E^{(N)}(g) = g \otimes \mathrm{id}$, which one can check commutes with the corresponding level structures q_{ι_1} and q_{ι_2} since $g \circ \iota_1 = \iota_2$.

When the underlying Drinfeld module is $\mathcal{E}_N^{(d)}$, we may also write lifts of the geometric Hecke operators $T_k^{(d)}(\mathfrak{p})$ to universal n -fold Drinfeld modules. Generalizing Φ and Ψ , we have morphisms $\Phi^\Lambda = \Phi_k^\Lambda(\mathfrak{p})$ and $\Psi^\Lambda = \Psi_k^\Lambda(\mathfrak{p}): \Lambda \otimes_A \mathcal{E}_{N,k,\mathfrak{p}}^{(d)} \rightarrow \Lambda \otimes_A \mathcal{E}_N^{(d)}$, where Φ lifts ϕ and is the identity on fibers, and Ψ lifts ψ and is given by $\Lambda \otimes_A E \rightarrow \Lambda \otimes_A E/C$ on fibers. (Here, we use that given an isogeny $E \rightarrow E'$ of Drinfeld modules, there is an obvious associated map of the fppf sheaves $\Lambda \otimes_A E \rightarrow \Lambda \otimes_A E'$ by tensoring with Λ on the level of functors of points.)

2.2 Motivic cohomology

For a smooth equidimensional scheme X over a field k , we have the bigraded motivic cohomology groups $H^\bullet(X, \mathbb{Z}(\bullet))$, which are contravariantly functorial, and covariantly functorial for proper maps, as defined for example in [MVW] (also a canonical reference for all the background that follows).

When $X = \mathrm{Spec} L$, then [Tot] proves that there is a natural isomorphism

$$H^n(X, \mathbb{Z}(n)) \xrightarrow{\sim} K_n^M(L)$$

between the degree- n , weight- n motivic cohomology and *Milnor K -theory*, which we recall is given in degree n by

$$K_n^M(L) := (L^\times \otimes \dots \otimes L^\times) / I$$

where I is the degree- n part of the ideal in the free tensor algebra on $L^\times$ generated by $x \otimes (1-x)$ for $x \in L^\times - \{0, 1\}$. When $n = 1$, there is more generally a map $H^1(X, \mathbb{Z}(1)) \rightarrow \Gamma(X, \mathcal{O}_X^\times)$ identifying the degree-1 weight-1 motivic cohomology with the global units of X . This allows us to more generally consider cup products like

$$u_1 \cup \dots \cup u_n \in H^n(X, \mathbb{Z}(n))$$

for units $u_1, \dots, u_n$ on X . Following notational conventions in Milnor K -theory, we will also denote such cup products, as well as the associated tensors in Milnor K -theory, by the curly braces $\{u_1, \dots, u_n\}$.

2.2.1 Tame symbols, functoriality, Suslin reciprocity

If $\iota: Z \hookrightarrow X$ is a closed immersion of smooth schemes of pure codimension d over a field, then there is a localization sequence

$$\dots \rightarrow H^{i-2d}(Z, \mathbb{Z}(n-d)) \rightarrow H^i(X, \mathbb{Z}(n)) \rightarrow H^i(X-Z, \mathbb{Z}(n)) \rightarrow H^{i+1-2d}(Z, \mathbb{Z}(n-d)) \rightarrow \dots$$

Such sequences are compatible with proper pushforward and flat pullback for pairs.

When $d = 1$, the boundary maps of the form $\partial: H^i(X-Z, \mathbb{Z}(n)) \rightarrow H^{i-1}(Z, \mathbb{Z}(n-1))$ are also called *tame symbols*: this name originates from maps in Milnor K -theory of the form

$$\partial: K_n^M(L) \rightarrow K_{n-1}^M(\ell)$$

where L is a field with a place ν with corresponding residue field ℓ ; this corresponds to the case $i = n$, Z is the point $\text{Spec } \ell$, and X is a scheme with function field L which has been localized at Z . In this setting, if π is a uniformizer at ν and $u_1, \dots, u_{n-1}$ are ν -units, then we have the formula

$$\partial\{\pi, u_1, \dots, u_{n-1}\} = \{u_1, \dots, u_{n-1}\},$$

and this formula actually fully specifies the map ∂ .

The last important property of tame symbols we will need is the *Suslin reciprocity law*, which was classically stated for tame symbols in Milnor K -theory. We will use the following “spread out” version of Suslin reciprocity for more general motivic cohomology (cf. [Kri] for the case that the base field is perfect).

Proposition 2.2.1. *Let $\pi: C \rightarrow B$ be a smooth projective morphism of dimension 1 between smooth varieties over a field, and let Z be a closed codimension 1 subscheme of C finite and dominant over every component of B . Then the composite*

$$H^p(C-Z, \mathbb{Z}(q)) \rightarrow H^{p-1}(Z, \mathbb{Z}(q-1)) \rightarrow H^{p-1}(B, \mathbb{Z}(q-1)) \quad (2.5)$$

of the residue and the pushforward is identically zero.

Proof. Since B , C , and Z are smooth over a field, Bloch's higher Chow groups compute motivic cohomology. Bloch's localization sequence for higher Chow groups gives a long exact sequence

$$\cdots \rightarrow H^p(C - Z, \mathbb{Z}(q)) \rightarrow H^{p-1}(Z, \mathbb{Z}(q-1)) \rightarrow H^{p+1}(C, \mathbb{Z}(q)) \rightarrow \cdots$$

where the first map is the residue map. The pushforward map $(\pi|_Z)_*$ factors through the pushforward by the inclusion $Z \hookrightarrow C$, which is the next arrow in the exact sequence. $\square$

2.3 Steinberg modules

Let V be an n -dimensional F -vector space.³ The *Steinberg module* $\text{St}(V)$, an infinite-dimensional representation of $\text{GL}_n(F)$, is the free module on symbols $[\ell_1, \dots, \ell_n]$ for lines $\ell_1, \dots, \ell_n \in \mathbb{P}(V)$ subject to the following relations:

- (degeneracy) $[\ell_1, \dots, \ell_n] = 0$ if the lines are not independent,
- (antisymmetry) $[\ell_1, \dots, \ell_n] = \text{sgn}(\sigma)[\ell_{\sigma(1)}, \dots, \ell_{\sigma(n)}]$ for any lines and any $\sigma \in S_n$,
- (Bykovskii) identifying any nonzero $v_1, \dots, v_n$ with their spanned lines,

$$[v_1, \dots, v_n] - [v_1 + v_2, v_2, \dots, v_n] + [v_1 + v_2, v_1, v_3, \dots, v_n] = 0.$$

Now suppose that we have a full A -lattice $V_0 \subset V$; let I be a nonzero proper ideal of A , and let $L \in \mathbb{P}(V_0/I)$. We write $\text{St}(V^L)$ for the submodule of $\text{St}(V)$ generated by symbols for which no sub- $(n-1)$ -tuple spans a hyperplane whose reduction modulo I contains L . This is the module which was denoted by ${}^L \text{St}(V)$ in [Xu2], where it is also proven [Xu2, Proposition 3.11]:

Proposition 2.3.1. *If $p \neq 2$, the same three above families of relations (degeneracy, antisymmetry, Bykovskii) on the smaller set of generators, given by tuples of lines whose proper subtuples do not span L modulo I , present $\text{St}(V^L)$.*

Also written in loc. cit. are resolutions, which we call “Whitney homology” or “Orlik–Solomon” resolutions, of $\text{St}(V)$ and $\text{St}(V^L)$, of the form

$$0 \rightarrow \text{St}(V) \rightarrow \bigoplus_{W \text{ codim } 1} \text{St}(W) \rightarrow \bigoplus_{W \text{ codim } 2} \text{St}(W) \rightarrow \cdots \rightarrow \bigoplus_{W \text{ dim } 1} \text{St}(W) \rightarrow \mathbb{Z} \rightarrow 0$$

³In practice, we may as well take a coordinatization $V = F^n$, but occasionally it is helpful to have a coordinate-free notation.

and

$$0 \rightarrow \mathrm{St}(V^L) \rightarrow \bigoplus_{W \text{ codim } 1}^I \mathrm{St}(W) \rightarrow \bigoplus_{W \text{ codim } 2}^I \mathrm{St}(W) \rightarrow \dots \rightarrow \bigoplus_{W \text{ dim } 1}^I \mathrm{St}(W) \rightarrow \mathbb{Z} \rightarrow 0$$

where the direct sums in the two resolutions go over all subspaces of the indicated dimension, respectively all subspaces of the indicated dimension which do not contain L modulo I . The latter resolution is a result of loc. cit. whereas the former dates much further back: e.g., see [OrSo] for the formulation of Orlik–Solomon, which also imparts an augmented cdga structure on the former resolution (simply given by $[\ell_1, \dots, \ell_j] \cdot [\ell_{j+1}, \dots, \ell_k] = [\ell_1, \dots, \ell_k]$). The differential in both resolutions is given by

$$[\ell_1, \dots, \ell_k] \mapsto \sum_{i=1}^k (-1)^i [\ell_1, \dots, \hat{\ell}_i, \dots, \ell_k]$$

for each $1 \leq k \leq n$, where for $k = 1$ this is just the augmentation map recording the total degree of the coefficients of line generators $[\ell]$ for the various dimension-1 subspaces.

Our main constructions will be arithmetic group-equivariant maps from $\mathrm{St}(V)$ and/or $\mathrm{St}(V^L)$ to various modules M of interest (which will be motivic cohomology groups), which we call “modular symbols” or “partial modular symbols” valued in M .

Remark 2.3.2. Taking the $\mathrm{GL}_n(F)$ -equivariant cohomology (respectively, equivariant cohomology for a subgroup with some congruence condition fixing the mod- I line L) of the Whitney homology resolutions yields “universal” $(n - 1)$ -cohomology classes for these groups with values in the Steinberg module, respectively its L -avoiding variant, relating our (partial) modular symbols to cocycles and cohomology like in the number field setting. We do not take this perspective in the present article.

Finally, we will also find the result [Xu2, Proposition 3.8] useful:

Proposition 2.3.3. *Let $\mathfrak{p}$ be a prime of A . Then any $\mathrm{St}(V)$ and $\mathrm{St}(V^L)$, as above, are also generated by the subset of their above-specified generating symbols $[\ell_1, \dots, \ell_n]$ which are unimodular at $\mathfrak{p}$, i.e. such that the reductions of the n lines modulo $\mathfrak{p}$ remain independent.*

2.3.1 The Steinberg module and Bruhat-Tits buildings

In the first author’s joint survey paper [FKS], a space of “cuspidal modular symbols” is defined in §4.1 in terms of *Bruhat–Tits buildings*, and conjectured to be the source of the Eisenstein maps whose construction is the aim of the present article. We review the definitions of these symbols here, to show how they are obtained from the Steinberg module formalism we employ in the rest of the article.

The Bruhat-Tits building $\mathcal{BT}_n$ for $\mathrm{PGL}_n(F_\infty)$ is the geometric realization of the $(n-1)$ -dimensional simplicial complex with i -simplices the equivalence classes of chains

$$\mathcal{L} = (L_1 \subsetneq L_2 \subsetneq \cdots \subsetneq L_{i+1})$$

of $\mathcal{O}_\infty$ -lattices L_i of rank n in F_∞^n , such that $\pi L_{i+1} \subsetneq L_1$, where $\mathcal{L}$ is equivalent to

$$(\pi L_{i+1} \subsetneq L_1 \subsetneq L_2 \subsetneq \cdots \subsetneq L_i)$$

and any scaling under $F_\infty^\times$, such that the faces of $\mathcal{L}$ are given by removing exactly one of the lattices from a chain defining $\mathcal{L}$.

For a basis $v_1, v_2, \dots, v_n$ of F_∞^n , we define the *apartment* $B = B_{v_1, \dots, v_n}$ in $\mathcal{BT}_n$ attached to $v_1, v_2, \dots, v_n$ to be the geometric realization of the largest simplicial subcomplex of the complex underlying $\mathcal{BT}_n$ with vertices $L_i = \sum_{j=1}^n \mathcal{O}_\infty \pi^{a_j} v_j$ for all $a_1, \dots, a_n \in \mathbb{Z}$. We say that an apartment B in $\mathcal{BT}_n$ is *rational* if the vectors $v_1, \dots, v_n$ that define it lie in F^n .

Remark 2.3.4. The apartment B can also be viewed as $X_*(T) \otimes_{\mathbb{Z}} \mathbb{R}$, where T is the torus in $\mathrm{PGL}_n(F_\infty)$ attached to any basis defining B , and $X_*(T) = \mathrm{Hom}(F_\infty^\times, T)$ is the cocharacter group. The normalizer $N = T \rtimes W$ of T in $\mathrm{PGL}_n(F_\infty)$, where $W \cong S_n$ is the Weyl group, then acts on B .

We have a canonical left action of $\mathrm{PGL}_n(F_\infty)$, and hence of $\mathrm{GL}_n(A)$, on $\mathcal{BT}_n$, induced by its usual action on lattices. Note that the class of the standard lattice $\mathcal{O}_\infty^n$ has stabilizer $\mathrm{PGL}_n(\mathcal{O}_\infty)$ in $\mathrm{PGL}_n(F_\infty)$.

In [FKS, §4.1], for an arithmetic subgroup $\Gamma \leq \mathrm{GL}_n(A)$, the authors call

$$\mathrm{im}(H_{n-1}(\Gamma \backslash \mathcal{BT}_n, \mathbb{Z}) \rightarrow H_{n-1}^{BM}(\Gamma \backslash \mathcal{BT}_n, \mathbb{Z})) \quad (2.6)$$

the space of “cuspidal modular symbols,” and speculate that it is the correct source module for Eisenstein maps. Note that in analogy with the classical setting of $\mathrm{GL}_2(\mathbb{Z})$ and modular curves, this is analogous to the parabolic (or “inner”) homology, i.e., the image of the homology of the open modular curve in its Borel–Moore homology. The full Borel–Moore homology can be identified, up to finite controllable error, with the full Steinberg module:

Theorem 2.3.5 (Theorem 12 of [KoYa2]). *The map $\mathrm{St}(n)_\Gamma \rightarrow H_{n-1}^{BM}(\Gamma \backslash \mathcal{BT}_n, \mathbb{Z})$ given by sending $[\ell_1, \dots, \ell_n]$ to the apartment $[B_{\ell_1, \dots, \ell_n}]$ is an inclusion with an explicitly bounded finite cokernel, which is p -typical if Γ has no prime-to- p torsion. More generally, the prime-to- p torsion always divides*

$$\prod_{i=1}^n (q_0^i - 1)$$

for q_0 the smallest residue characteristic of any non-infinite place of F .

Our Eisenstein maps are actually definable on the entirety of $\mathrm{St}(n)_\Gamma$, i.e., essentially (up to highly controlled finite index) the whole Borel–Moore homology of the Bruhat–Tits building. However, desired Hecke equivariance and integrality properties are only expected to hold on submodules given by the images of maps

$$\mathrm{St}(n^L)_\Gamma \rightarrow \mathrm{St}(n)_\Gamma \tag{2.7}$$

for certain Γ -stable sets of lines L , and this is what we will actually prove. Generally speaking, our strategy will be to construct the maps and prove Hecke/integrality properties on the modules $\mathrm{St}(n^L)_\Gamma$; we will then prove that the maps factor through the image of (2.7), and view this as an “extra parabolicity” result (for the “cusps”/ends of the Bruhat–Tits building excluded by the L -avoiding condition).

For any proper subset of lines L , we expect that there is some analogue [KoYa2] “relative to a partial boundary” of the Bruhat–Tits building. This morally corresponds to taking “partial compactifications” of the building associated to a subset of parabolic subgroups of GL_n , but the optimal results in loc. cit. are not actually obtained using compactifications and it is not clear whether this moral story should be literally true in the function field setting (as it is for archimedean symmetric spaces for number fields). Thus, there is no obvious way to cleanly write down a statement involving relative cohomology for some boundary component. We therefore do not pursue this formulation, instead simply defining our Eisenstein modular symbols as maps from the image of (2.7).

3 The cocycle and its properties

3.1 Constructing a modular symbol

With the preliminaries out of the way, we now turn to constructing a modular symbol (and hence a cocycle) valued in the motivic cohomology of an n -fold Drinfeld module $T := T_E(\Lambda)$, for some $\Lambda \subset F^n$ and Drinfeld module $E \rightarrow S$. As mentioned previously, the integral matrix algebra $\mathrm{End}_A(\Lambda) \subset M_n(F)$ has a natural left action on this space and hence a left pushforward action on its motivic cohomology for the subgroup $\mathrm{Aut}_A(\Lambda)$. However, our modular symbols will actually be equivariant for the action of a much larger S -arithmetic subgroup. For this, we need the following notion.

Definition 3.1.1. Write $M^{(0)}$ for the *trace-fixed* submodule of an A -module M ; that is, the submodule of elements $m \in M$ such that $[a]_* m = m$ for all $a \in A - \{0\}$.

Then we can define a $\mathrm{GL}_n(F)$ -action on the trace-fixed part of the motivic cohomology groups $H^\bullet(T, \mathbb{Z}(\bullet))^{(0)}$ by having a matrix γ act by $[a \cdot \gamma]_*$ for any $a \in A$ such that $\gamma \in \mathrm{End}_A(\Lambda)$, which is well-defined by the trace-fixedness. From this, we see these groups in fact carry a $\mathrm{PGL}_n(F)$ -action, but we will think of it as a $\mathrm{GL}_n(F)$ -action trivial on the center.

Let us now restrict attention to the case $\Lambda = A^n$, so that $T = E_1 \times_S \dots \times_S E_n$; here we write subscripts only to keep track of the coordinates (though this means that our discussion will actually also apply to all “diagonal” lattices Λ as well). Then via the theta element construction of Section 2.1, we have a canonical element $\theta_{E_i} \in (\mathcal{O}(E_i - S)^\times \otimes \mathbb{Z}')^{(0)}$. Writing $\pi_i: T \rightarrow E_i$ for the projection to the i th factor, we obtain functions $\theta_i := \pi_i^* \theta_{E_i} \in \mathcal{O}(T - \pi_i^{-1}(S))^\times \otimes \mathbb{Z}'$, for $1 \leq i \leq n$. Each individual θ_i is no longer trace-fixed. However, their cup product

$$\theta_1 \cup \dots \cup \theta_n \in H^n \left(T - \bigcup_i \pi_i^{-1}(S), \mathbb{Z}'(n) \right)^{(0)}$$

does have this property: we can break up $[a]_*$ into the composite of the pushforwards by the n diagonal matrices $\mathrm{diag}(1, \dots, 1, a, 1, \dots, 1)$, each of which fixes $\theta_1 \cup \dots \cup \theta_n$ by the push-pull formula.

We are now ready to state the main construction:

Theorem 3.1.2. *Suppose $p \neq 2$. Let E be a Drinfeld A -module and Λ a full lattice in F^n ; let $T = T_E(\Lambda)$ as before. Then there is an injective modular symbol over F given by the $\mathrm{GL}_n(F)$ -equivariant map*

$$\Theta_T: \mathrm{St}(\Lambda) \hookrightarrow K_n^M(k(T))^{(0)} \otimes \mathbb{Z}'$$

such that when $T = E^n$, it is given by

$$[\ell_1, \ell_2, \dots, \ell_n] \mapsto \begin{pmatrix} \ell_1 & \dots & \ell_n \end{pmatrix}_* \{\theta_1, \dots, \theta_n\} \quad (3.1)$$

where here the columns of the matrix indicate that we can choose any nonzero vector on the line $\ell_\bullet$. Furthermore, if $g: \Lambda \rightarrow \Lambda'$ is a morphism of lattices, and $\varphi_g: T \rightarrow T'$ the associated isogeny of n -fold Drinfeld modules, we have $(\varphi_g)_* \Theta_T(\sigma) = \Theta_{T'}(g \cdot \sigma)$.

Proof. We claim that the construction and claims of the theorem for general T follow immediately from the case $T = E^n$ (for any Drinfeld module E). Indeed, there is an ideal isogeny

$$\varphi_g: E^n \rightarrow T$$

for some suitable E and some $g \in \mathrm{GL}_n(F)$ such that $gA^n \subset \Lambda$. Then assuming the existence of Θ_{E^n} with the given properties, we claim that we simply *define*

$$\Theta_T := ((\varphi_g)_* \Theta_{E^n}) \circ g^{-1}.$$

The map $(\varphi_g)_*$ is invertible on trace-fixed elements, so Θ_T inherits the injectivity of Θ_{E^n} as well.

It is easy to check that the above definition implies similar functoriality between arbitrary isogenies $T \rightarrow T'$; thus, it remains only to show well-definedness: i.e., when $\varphi_g: T \rightarrow T$ is a self-isogeny, that $\Theta_T = ((\varphi_g)_* \Theta_T) \circ g^{-1}$. This again formally reduces to checking the case $T = E^n$, in which case the formula (3.1) shows that we only need to check

$$\begin{pmatrix} \ell_1 & \dots & \ell_n \end{pmatrix}_* \{\theta_1, \dots, \theta_n\} = (\varphi_g)_* \Theta_{E^n}(g^{-1}[\ell_1, \dots, \ell_n])$$

which is clear, since $\varphi_g: T \rightarrow T$ in this case is simply the map given by the matrix g viewed in $\text{End}_A(E^n)$ in the usual way.

We thus restrict ourselves to $T = E^n$. We first note that the claim that the choice of representative vector in (3.1) does not change the result is a consequence of the trace-fixedness of each θ_i ; i.e., $\{\theta_1, \dots, \theta_n\}$ is fixed under pushforward by diagonal matrices.

Then we claim that (3.1) provides a well-defined map on $\text{St}(n)$, which is to say that the Steinberg relations hold between the elements of Milnor K -theory. The $\text{GL}_n(F)$ -equivariance of the resulting map is visible from the formula.

The antisymmetry and degeneracy relations of the Steinberg module are formal; the only one that needs checking is the Bykovskii relation. We prove it in the Bykovskii formulation, i.e., that

$$\sum_{i=1}^{n+1} (-1)^{i-1} [\ell_1, \dots, \widehat{\ell_i}, \dots, \ell_{n+1}] = 0 \quad (3.2)$$

for any tuple of lines $\ell_1, \dots, \ell_{n+1}$ spanning all of F^n such that the only linear dependency is that ℓ_1, ℓ_2, ℓ_3 lie in the same plane. By moving around relations using the $\text{GL}_n(F)$ action, this reduces to the case of the Bykovskii relation for $e_1, \dots, e_n, e_1 + e_2$. The resulting pushforward relation is actually just the cup product of the pullback of the relation for $n = 2$ with $\theta_3 \cup \dots \cup \theta_n$, so we are reduced to checking this case. Then our desired relation can be spelled out explicitly as

$$\{\theta_1, \theta_2\} - \{\theta(t_1 - t_2), \theta_2\} + \{\theta(t_2 - t_1), \theta_1\} = 0.$$

Here, we write t_1 and t_2 for the fixed local coordinate on E viewed in the first and second slots of E^2 ; the notation we use is only suggestive, since these coordinates may not be globally defined over S . Rigorously, what we mean by the functions $\theta(t_1 - t_2)$ and $\theta(t_2 - t_1)$ are the pullbacks of θ from the subtraction map $E^2 \rightarrow E$ and its negative.

To prove this, we apply Proposition 2.2.1 with $B = E^2 - E \times \{0\} - \{0\} \times E - \Delta$ for Δ the diagonal. The family C we then take to be the projectivization of the line bundle $E \times_S B \rightarrow B$.

If we write t for the same local coordinate on the new factor of E , we let Z then be the locus $\{t = 0\} \cup \{t = t_1\} \cup \{t = t_2\} \cup \{t = \infty\}$, and consider the element

$$\{\theta(t_1 - t), \theta(t_2 - t), \theta(t)\} \in H^3(C - Z, \mathbb{Z}'(3)).$$

Then Suslin reciprocity says that the tame symbol to Z and pushdown to B of this vanishes; we compute that this is precisely

$$\{\theta_1, \theta_2\} - \{\theta(t_1 - t_2), \theta_2\} + \{\theta(t_2 - t_1), \theta_1\} + (\pi_\infty)_* \partial_{t=\infty} \{\theta(t_1 - t), \theta(t_2 - t), \theta(t)\}$$

where π_∞ is the map pushing the $t = \infty$ locus down to B . We thus are reduced to showing that the boundary-at-infinity contribution vanishes. Indeed, if we rewrite the degree-3 element as

$$\{\theta(t_1 - t)/\theta(t), \theta(t_2 - t)/\theta(t), \theta(t)\}$$

we observe from the explicit formula (2.2) that the former two terms both extend regularly over this boundary term, with regularized value 1, while $\theta(t)$ is an (inverse) uniformizer at this boundary. Hence the tame symbol vanishes, as desired.

The only thing that remains to prove is that Θ_T is injective. Observe that its composition with the tame symbol fits into a commutative diagram

$$\begin{array}{ccc} \text{St}(n) & \xrightarrow{\Theta_T} & K_n^M(k(T))^{(0)} \otimes \mathbb{Z}' \xrightarrow{\partial} \bigoplus_{x \in T^{(1)}} K_{n-1}^M(k(x))^{(0)} \otimes \mathbb{Z}' \\ \downarrow \partial & & \parallel \\ \bigoplus_{\substack{V \subseteq F^n \\ \text{codim } 1}} \text{St}(V \cap \Lambda) & \xrightarrow{\bigoplus_V \Theta_V} & \bigoplus_{x \in T^{(1)}} K_{n-1}^M(k(x))^{(0)} \otimes \mathbb{Z}' \end{array} \quad (3.3)$$

where the V -indexed realization map Θ_V is short for the map $\Theta_{T_E(V \cap \Lambda)}: \text{St}(V \cap \Lambda) \rightarrow K_{n-1}^M(k(T_E(V \cap \Lambda)))$. Here, $T_E(V \cap \Lambda)$ is viewed as a subscheme over S of $T_E(\Lambda)$ in the natural way, by including $(V \cap \Lambda) \otimes_A E \hookrightarrow \Lambda \otimes_A E$ on the level of functors of points.

Then the result follows by the exactness of the Orlik-Solomon complex (i.e., the injectivity in top degree), and induction on n . $\square$

Remark 3.1.3. The idea of using Suslin reciprocity to prove relations of this form was inspired by the article [BPPS]. The previous approach of the second author using cubical Bloch cycles (as employed in, e.g., [Xu] for algebraic tori) runs into technical obstacles from impropriety of intersection with faces.

Remark 3.1.4. The preceding proof can be used to construct a realization map from the full Orlik-Solomon complex to the Gersten complex. If one replaces the Orlik-Solomon complex

by a matroid complex, one can even map to a motivic complex and relate the resulting modular symbol to an equivariant polylogarithm class. In the Drinfeld setting, we do not presently see any additional benefit arising from this formulation, so we omit it from this article. (Simple abelian schemes of dimension greater than 1 provide such motivation in the number field setting, as it is unknown how to define their equivariant polylogarithm classes by explicit cup products of units; this has no obvious analogue in Drinfeld modules.)

3.2 Pulled back cocycles

3.2.1 Avoiding the bad locus

We first remark that thanks to Proposition 2.3.1, the proof of Theorem 3.1.2 works not only for the entirety of $\text{St}(\Lambda)$, but also for our restricted submodules: fix a nonzero ideal $N \subset A$, and take $L = L_N$ to be the set of lines reducing to $[0 : \dots : 0 : 1]$ modulo N (that is, the line corresponding to the factor E_n of T).

Theorem 3.2.1. *Let $\Gamma \leq \text{GL}(\Lambda \otimes A_{(N)})$ be the subgroup which stabilizes the level structure L_N , and $T = T_E(\Lambda)$. Then there is a Γ -equivariant “partial” modular symbol*

$$\Theta_T : \text{St}(\Lambda^L) \rightarrow H^n(T - Z(L), \mathbb{Z}'(n))^{(0)}$$

given by the same formula (3.1) in the case $\Lambda = A^n$ (and satisfying analogous functoriality), where $Z(L)$ is the hyperplane arrangement inside T corresponding to the kernels of the dual bases of elements in $\text{St}(\Lambda^L)$, identifying subspaces of T with subspaces of V^n in the natural way.

This version of the cocycle is now “spread out” from the generic point, allowing us to pull back by Γ -fixed torsion cycles in the N -torsion line corresponding to $[0 : \dots : 1]$. In theory, there are many possible choices of subgroup Γ and torsion sections to pull back by, but in the present article we will choose them informed by the approach of [ShVe], and in particular applications to the first author’s conjectures relating K -theory and the homology of arithmetic groups.

Let us give the simplest example of the kind of resulting specializations we will consider, when $T = E^n$: let $Y = Y_N^{(d)}$ over U_N , with its canonical N -level structure

$$\iota_N^{(d)} : Y \rightarrow E = \mathcal{E}_N^{(d)}$$

distinguishing an N -torsion section of E . Consider then the section

$$s : Y \rightarrow T, y \mapsto (0, \dots, 0, \iota_N^{(d)}(y)).$$

We define an arithmetic subgroup

$$\Gamma_0^{(n)}(N) = \text{Stab}([0 : 0 : \dots : 0 : 1] \in \mathbb{P}^{n-1}(A/N)) \subset \text{GL}_n(A).$$

Note that s is *not* fixed by $\Gamma_0^{(n)}(N)$, in the sense that $s \circ \gamma$ is not again the section s , but rather $[a] \circ s$ for some association $\gamma \mapsto a \in (A/N)^\times$. Writing

$$\langle - \rangle : \Gamma_0^{(n)}(N) \rightarrow (A/N)^\times$$

for this homomorphism with kernel $\Gamma_1^{(n)}(N)$, we observe that this implies that s is a $\Gamma_0^{(n)}(N)$ -equivariant map for the diamond action $\gamma \cdot x := \langle \gamma \rangle x$ of $\Gamma_0^{(n)}(N)$ on Y . For short, for any T with its distinguished section s , we will write $H(T; N)$ for the direct limit of the groups $H^n(U, \mathbb{Z}'(n))$ over open subschemes $U \subset T$ containing the images of $[a] \circ s$ for all $a \in (A/N)^\times$, and $H(Y) := H^n(Y, \mathbb{Z}'(n))$. Then we have a $\Gamma_0^{(n)}(N)$ -equivariant pullback map

$$s^* : H(T; N) \rightarrow H(Y)$$

and consequently a pulled-back “partial modular symbol”

$$s^* \Theta_T : \text{St}((A^n)^{L_N}) \rightarrow H(Y)$$

In general, we will use the level- N functor $T_E^{(N)}$ defined in (2.4) to define our specialization section. The above example corresponds to $\Lambda = A^n$, $\Lambda' = A^{n-1} \times N^{-1}A$, and ι the obvious inclusion in the last coordinate.

To formulate the general case, let $\mathcal{X} := (\Lambda, \iota : N^{-1}A/A \rightarrow \Lambda'/\Lambda)$ be any object of $\text{Lat}_N(n)$, and $(T_E(\Lambda), q)$ its image under $T_E^{(N)}$; let $\Gamma_1(\mathcal{X})$ be the stabilizer of $\mathcal{X}$ inside $\text{GL}_n(F)$, and $\Gamma_0(\mathcal{X})$ the stabilizer of the $\text{Aut}_A(N^{-1}A/A)$ -orbit of $\mathcal{X}$. We then construct, for $T = T_E(\Lambda)$ now, the section

$$s_{\mathcal{X}} : Y \rightarrow T, y \mapsto q(\iota_N^{(d)}(y))$$

which is again $\Gamma_0(\mathcal{X})$ -equivariant for the diamond action of $\Gamma_0(\mathcal{X})/\Gamma_1(\mathcal{X}) \cong \text{Aut}_A(N^{-1}A/A) \cong (A/N)^\times$ on Y , and consequently an equivariant (for the same action) pullback

$$s_{\mathcal{X}}^* : H(T; N) \rightarrow H(Y)$$

resulting in $s^* \Theta_T : \text{St}(\Lambda^{L_N}) \rightarrow H(Y)$ as before.

In practice, we will sometimes also consider the above construction on a concrete set of representatives, rather than arbitrary lattices with level structure: recall that we have the finite adelic space of level $\Gamma_1(N)$

$$\text{GL}_n(F) \backslash \text{GL}_n(\mathbb{A}_F^f) / \hat{\Gamma}_1^{(n)}(N) =: S_{\Gamma_1(N)}^{(n),f} \xrightarrow{\det} S_1^{(1),f} = \text{Pic}(A),$$

which we can interpret as the set of isomorphism classes of objects in $\mathbf{Lat}_N(n)$. Pick a skeleton of this category; on objects, this amounts to a set of representatives $\{\mathcal{X}_c\}_c := \{(X_c, \iota_c)\}_c$ for the finite adelic space, indexed by the corresponding class $c \in \text{Pic}(A)$. For each (X_c, ι_c) , we write $\Gamma_1^{(c)}$ for $\text{Aut}(\mathcal{X}_c)$, $\Gamma_0^{(c)}$ for the stabilizer of the $\text{Aut}_A(N^{-1}A/A)$ -orbit of $\mathcal{X}_c$, and Δ_c for $\text{End}(\mathcal{X}_c)$. We write also $\Delta_{b,c}$ for the hom-space $\text{Hom}(\mathcal{X}_b, \mathcal{X}_c)$, which is also a subset of $\text{GL}_n(F)$. We then deduce partial modular symbols of the form

$$s_c^* \Theta_c := s_c^* \Theta_{T_c} : \text{St}(X_c^{L_N}) \rightarrow H(Y)$$

as above, $\Gamma_0^{(c)}$ -equivariant for the diamond-twisted action of $\Gamma_0^{(c)}/\Gamma_1^{(c)} \cong (A/N)^\times$ on the target as before.

4 Hecke actions

In this section, we will define “cohomological” Hecke operators which act on the modular symbols (or resulting cocycles) of the form $s_{\mathcal{X}}^* \Theta_{T_E(\mathcal{X})}$. These contrast with the earlier-defined “geometric” Hecke operators for GL_d acting on Drinfeld modular varieties, and the universal modules above them. Our main theorem will show that in this latter setting, our Eisenstein symbol/cocycle satisfies a compatibility property between the two types of operators. This compatibility is of a particularly simple and interesting form when $d = n$ or $d = n - 1$, where it yields an equivariance statement, respectively one of the two maps (usually called ϖ_N) predicted by the so-called Sharifi conjectures.

We will generally follow the approach of [ShVe, 6.5] to comparing a pair of Hecke actions, though with the key differences that we define the cohomological Hecke action on the level of modular symbols rather than cocycles, and their cocycles are written with pullback notation, while ours are written with pushforward (though there will be pullback maps involved in the *definitions* of the Hecke actions). To be more precise, for the Hecke formalism itself, we will follow the more general approach in the later article [LSSW], in order to account for non-triviality of the class group (though in that article the theory of “geometric” Hecke operators reduces to a Galois action, since it is effectively in the $d = 1$ setting).

4.1 Cohomological Hecke action

Let us fix our setup for cohomological Hecke operators on modular symbols. Our goal will be a description which works both on the level of families of n -dimensional Drinfeld modules, as well as specialized to the Drinfeld modular variety via a torsion section fixed by some

level structure; thus, we will need a formalism which “respects an N -torsion section” for this compatibility to be maintained. We will use the adelic monoid $\hat{\Delta}_N^{(n)}$, which recall is defined as the stabilizer of $\bar{N} \subset (F/A)^n$ in $M_n(\hat{A}) \cap \mathrm{GL}_n(\mathbb{A}_F^f)$. In this subsection, we take $n \geq 2$ and restrict to double cosets whose determinants are units at all primes dividing N , so that the resulting maps also preserve our restricted Steinberg modules.

Write $\mathbf{Lat}_N(n)^{\mathrm{iso}}$ for the groupoid with the same objects as $\mathbf{Lat}_N(n)$ and only its isomorphisms as morphisms. To be more precise, we consider

$$\mathcal{M}: \mathcal{X} \mapsto \mathrm{Hom}_{\mathrm{Aut}(\mathcal{X})}(\mathrm{St}(\Lambda^{L_N}), H(T_E(\Lambda); N)), \quad s\mathcal{M}: \mathcal{X} \mapsto \mathrm{Hom}_{\mathrm{Aut}(\mathcal{X})}(\mathrm{St}(\Lambda^{L_N}), H(Y))$$

as presheaves on $\mathbf{Lat}_N(n)^{\mathrm{iso}}$ with a natural transformation between them given by the pullback $s_{\mathcal{X}}^*: \mathcal{M} \rightarrow s\mathcal{M}$. Here, associated to an isomorphism $\mathrm{GL}_n(F) \ni h: \mathcal{X} \rightarrow \mathcal{X}'$, we have the associated structure maps

$$h^*: \mathrm{Hom}_{\mathrm{Aut}(\mathcal{X}')}(\mathrm{St}(\Lambda'^{L_N}), H(T_E(\Lambda'); N)) \rightarrow \mathrm{Hom}_{\mathrm{Aut}(\mathcal{X})}(\mathrm{St}(\Lambda^{L_N}), H(T_E(\Lambda); N))$$

sending $\theta \mapsto h^* \circ \theta \circ h_*$, and similarly for $s\mathcal{M}$ (noting that the outside pullback h^* becomes trivial). Observe that in particular, if h is an *automorphism* of $\mathcal{X}$, these presheaf structure maps are the identity by $\mathrm{Aut}(\mathcal{X})$ -equivariance.

Then Theorem 3.1.2 (or rather its “spread out version”) shows that $\mathcal{X} \mapsto \Theta_{T_E(\Lambda)}$ and $\mathcal{X} \mapsto s_{\mathcal{X}}^* \Theta_{T_E(\Lambda)}$ are global sections of these presheaves. Their compatibility here is only with isomorphisms; we will use the additional pushforward compatibility of Θ with isogenies to compute its Hecke action below.

There is an action of the whole adelic group $\mathrm{GL}_n(\mathbb{A}_F^f)$ on the objects of $\mathbf{Lat}_N(n)$, given on an adelic matrix g by sending a lattice Λ to

$$g \cdot \Lambda := F^n \cap g(\hat{A} \otimes_A \Lambda).$$

This extends to the N -level structure by identically translating the auxiliary lattice Λ' to $g \cdot \Lambda'$, and sending the trivialization ι to the composite

$$g \circ \iota: N^{-1}A/A \xrightarrow{\iota} \Lambda'/\Lambda \xrightarrow{g} g\Lambda'/g\Lambda,$$

where the last map is induced on the quotients of the completed lattices. For any object $\mathcal{X}$, write $\hat{\Gamma}_{\mathcal{X}}$ for its adelic stabilizer, including the trivialization ι , and write $\hat{\Delta}_{\mathcal{X}}$ for the monoid of adelic matrices g for which the identity matrix in $\mathrm{GL}_n(F)$ is a map from $g \cdot \mathcal{X}$ to $\mathcal{X}$. In particular, integrality here is relative to $\hat{A} \otimes_A \Lambda$.

Let $\mathcal{X}_0 = (\Lambda_0, \iota_0)$ be given by $\Lambda_0 = A^n$, $\Lambda'_0 = A^{n-1} \times N^{-1}A$ and the obvious map $\iota_0: N^{-1}A/A \rightarrow \bar{N} = \Lambda'_0/\Lambda_0$; then $\hat{\Delta}_{\mathcal{X}_0} = \hat{\Delta}_N^{(n)}$ and $\hat{\Gamma}_{\mathcal{X}_0} = \hat{\Gamma}_1^{(n)}(N)$ as previously defined.

One further has *canonical* identifications of adelic double coset spaces

$$\hat{\Gamma}_1^{(n)}(N) \backslash \hat{\Delta}_N^{(n)} / \hat{\Gamma}_1^{(n)}(N) \xrightarrow{\sim} \hat{\Gamma}_{\mathcal{X}} \backslash \hat{\Delta}_{\mathcal{X}} / \hat{\Gamma}_{\mathcal{X}}, \quad (4.1)$$

sending the class of g to the class of $t_{0 \rightarrow \mathcal{X}} g t_{\mathcal{X} \rightarrow 0}$, for $t_{0 \rightarrow \mathcal{X}}$ sending $\mathcal{X}_0$ to $\mathcal{X}$ and $t_{\mathcal{X} \rightarrow 0} = t_{0 \rightarrow \mathcal{X}}^{-1}$. Changing $t_{0 \rightarrow \mathcal{X}}$ amounts to right multiplication by an element of $\hat{\Gamma}_1^{(n)}(N)$, so the identification is independent of this choice.

For each fixed object $\mathcal{X}$ and double coset T_g , view g as an element of $\hat{\Delta}_{\mathcal{X}}$ via (4.1), and decompose

$$\hat{\Gamma}_{\mathcal{X}} g \hat{\Gamma}_{\mathcal{X}} = \bigsqcup_{g_i \in I_g} \hat{\Gamma}_{\mathcal{X}} g_i.$$

The inverse translates $\mathcal{X}_i = g_i^{-1} \cdot \mathcal{X}$ are a finite set of objects independent of the representatives. Their underlying lattices contain Λ , and we have natural inclusion morphisms

$$1_i: \mathcal{X} \rightarrow \mathcal{X}_i.$$

Indeed, g_i fixes the level structure on $\mathcal{X}$, so the quotient map $F^n / \Lambda \rightarrow F^n / \Lambda_i$ sends ι to $g_i^{-1} \circ \iota$. The lattices Λ and Λ_i agree at all primes dividing N , and thus their restricted Steinberg modules are identified by the identity on F^n . We use 1_i also for the associated isogeny $T_E(\Lambda) \rightarrow T_E(\Lambda_i)$. (Note there is a dependence on $\mathcal{X}$ which we are suppressing in the notation.)

For $\mathcal{F} = \mathcal{M}$ or $s\mathcal{M}$, we may now define operators on global sections by

$$T_g \sigma := \left(\mathcal{X} \mapsto \sum_{g_i \in I_g} 1_i^* \circ \sigma(\mathcal{X}_i) \circ (1_i)_* \right),$$

where the rightmost map is the identity on the restricted Steinberg module; for $s\mathcal{M}$, the outside pullback is the identity on $H(Y)$. The pullbacks on motivic cohomology are taken in the direct limits defining $H(T; N)$. Each summand is defined before taking $\text{Aut}(\mathcal{X})$ -invariants. Since $\text{Aut}(\mathcal{X})$ permutes the inverse translates, their sum is invariant, and one checks in the same way that it furnishes a global section independent of all choices.

With the multiplication convention of Section 2.1.5, this is a right Hecke action: although we write the operators on the left, $T_h(T_g \sigma) = (T_g T_h) \sigma$. Indeed, composing the inverse translates gives $(g_i h_j)^{-1} \cdot \mathcal{X}$, and their multiplicities are the coefficients of $T_g T_h$. On the commutative unramified Hecke algebra used below, this also gives a left action.

For the section Θ , Theorem 3.1.2 gives

$$\Theta_{T_E(\Lambda_i)} \circ (1_i)_* = (1_i)_* \circ \Theta_{T_E(\Lambda)},$$

where the two pushforwards act on the Steinberg module and motivic cohomology, respectively. Consequently,

$$(T_g \Theta)_{T_E(\Lambda)} = \left(\sum_{i \in I_g} 1_i^* (1_i)_* \right) \circ \Theta_{T_E(\Lambda)}.$$

Moreover, $1_i \circ s_{\mathcal{X}} = s_{\mathcal{X}_i}$, so the natural transformation $s^*: \mathcal{M} \rightarrow s\mathcal{M}$ is equivariant for the Hecke action, term by term. This applies in particular to the operators $T_k^{(n)}(\mathfrak{p}^m)$ for $\mathfrak{p} \nmid N$ defined in Section 2.1.5.

We conclude by translating our adelic Hecke operators into the language of Δ -module systems defined in [LSSW] (building on the earlier Δ -modules of [ShVe]); the “ Δ ”s in question come from the earlier-defined monoids Δ_c and hom-sets $\Delta_{b,c}$, restricted here to isogenies invertible at the primes dividing N . Pullback acts on the coefficient groups before taking automorphism invariants. Because automorphisms act trivially on the structure maps of $\mathcal{M}$ and $s\mathcal{M}$, their values are *canonically* identified over any isomorphism class, so their global sections can be described in terms of our fixed skeletal representatives $\mathcal{X}_1, \dots, \mathcal{X}_h$.

Proposition 4.1.1. *For any global section σ of $\mathcal{M}$ or $s\mathcal{M}$ on $\mathbf{Lat}_N(n)^{\text{iso}}$,*

$$(T_g \sigma)(\mathcal{X}_b) = \sum_{h_i \in I_{b,c}} h_i^* \circ \sigma(\mathcal{X}_c) \circ (h_i)_*.$$

Here $c = b[\mathfrak{a}_g]^{-1}$, where $\mathfrak{a}_g = (\det g)\hat{A} \cap F$ is the ideal attached to $\det g$. If t_b, t_c send $\mathcal{X}_0$ to $\mathcal{X}_b, \mathcal{X}_c$, respectively, then $I_{b,c}$ is a set of representatives for

$$\Gamma_1^{(c)} \backslash \left(\text{GL}_n(F) \cap t_c \hat{\Gamma}_1^{(n)}(N) g \hat{\Gamma}_1^{(n)}(N) t_b^{-1} \right).$$

In particular, $h_i \in \Delta_{b,c}$, and all these matrices are rational.

Proof. The inverse translates $\mathcal{X}_i$ of $\mathcal{X}_b$ have determinant class $b[\mathfrak{a}_g]^{-1}$, so choose isomorphisms $\gamma_i: \mathcal{X}_i \xrightarrow{\sim} \mathcal{X}_c$. Set $h_i = \gamma_i \circ 1_i$. Compatibility of σ with these isomorphisms allows us to rewrite the summands as

$$1_i^* \circ \sigma(\mathcal{X}_i) \circ (1_i)_* = h_i^* \circ \sigma(\mathcal{X}_c) \circ (h_i)_*.$$

Changing γ_i multiplies h_i on the left by $\Gamma_1^{(c)}$. Conversely, a matrix in the displayed adelic double coset gives the inverse translate $h_i^{-1} \mathcal{X}_c$, and two such matrices give the same object exactly when they differ by left multiplication by $\Gamma_1^{(c)}$. The stated description of $I_{b,c}$ and the formula follow. $\square$

Remark 4.1.2. The Hecke action on a set of skeletal representatives determines its action on the whole presheaf $\mathcal{F} = \mathcal{M}$ or $s\mathcal{M}$ on $\mathbf{Lat}_N(n)^{\text{iso}}$, in the following precise sense: if we have some isomorphism $\gamma: \mathcal{X}_c \xrightarrow{\sim} \mathcal{X}$ for $\gamma \in \text{GL}_n(F)$, then for any Hecke operator T_g and global section σ of $\mathcal{F}$,

$$\gamma^*(T_g\sigma)(\mathcal{X}) = (T_g\sigma)(\mathcal{X}_c),$$

where γ^* denotes the presheaf structure map. This follows directly from unraveling definitions.

4.1.1 fppf-local capitulation of the cohomological Hecke action

Suppose now that $\mathfrak{p} \nmid N$ and all the $\mathfrak{p}$ -torsion of E has a Drinfeld trivialization over S (which recall is always true fppf-locally). Then we may label sections by the trivialization map $\mathfrak{p}^{-1}A^d/A^d \rightarrow E[\mathfrak{p}]$, and similarly by

$$\Lambda \otimes_A (\mathfrak{p}^{-1}A^d/A^d) \cong \mathfrak{p}^{-1}\Lambda^d/\Lambda^d \rightarrow T_E(\Lambda)[\mathfrak{p}]$$

for any $\Lambda \in \mathbf{Lat}(n)$. For example, explicitly, we may define $Y_{N,\bar{\mathfrak{p}}}^{(d)}$ to be the Drinfeld modular variety (over U_N or $\text{Spec } A$, as usual) corresponding to Drinfeld level structure

$$\mathcal{L} = \bar{N} + (\mathfrak{p}^{-1}A/A)^d \subset (F/A)^d,$$

which represents a moduli problem with a full Drinfeld trivialization of the $\mathfrak{p}$ -torsion, in addition to the original data coming from $\bar{N}$. Then

$$\pi_{\mathfrak{p}}: Y_{N,\bar{\mathfrak{p}}}^{(d)} \rightarrow Y_N^{(d)}$$

is an fppf covering.

Given $x \in (\mathfrak{p}^{-1}A/A)^d$, write

$$t_x: \mathcal{E}_{N,\bar{\mathfrak{p}}} \rightarrow \mathcal{E}_{N,\bar{\mathfrak{p}}}$$

for the “translation-by- $\iota(x)$ ” map, defined in the usual way: i.e., as the correspondence

$$\mathcal{E}_{N,\bar{\mathfrak{p}}} \xrightarrow{(\text{id}, \iota(x) \circ \text{projection})} \mathcal{E}_{N,\bar{\mathfrak{p}}}^2 \rightarrow \mathcal{E}_{N,\bar{\mathfrak{p}}}$$

where the last arrow is the group scheme operation. More generally, we may use the same notation for $x \in \Lambda \otimes_A (\mathfrak{p}^{-1}A/A)^d$ for translations on $T_E(\Lambda)$.

Write $I_k(\mathfrak{p})$ for the index set of the coset decomposition for $T_k^{(n)}(\mathfrak{p})$. The associated inverse translates Λ_i are precisely the lattices satisfying

$$\Lambda \subset \Lambda_i \subset \mathfrak{p}^{-1}\Lambda, \quad \dim_{A/\mathfrak{p}}(\Lambda_i/\Lambda) = k,$$

with level structure induced from $\mathcal{X}$. Equivalently, these are indexed by the k -planes $H_i = \mathfrak{p}\Lambda_i/\mathfrak{p}\Lambda$ in $\Lambda/\mathfrak{p}\Lambda$. Then we have the following result:

Proposition 4.1.3. *Assuming all the $\mathfrak{p}$ -torsion of E has a Drinfeld trivialization over its base S , we have the equality of correspondences on $T_E(\Lambda) \times_S T_E(\Lambda)$*

$$\sum_{i \in I_k(\mathfrak{p})} 1_i^*(1_i)_* = \sum_{H \in \text{Gr}(\Lambda/\mathfrak{p}\Lambda, k)} \sum_{x \in H \otimes_{A/\mathfrak{p}} (\mathfrak{p}^{-1}A^d/A^d)} t_x^*$$

for any $\mathfrak{p} \nmid N$ and $0 \leq k \leq n$. Here, $\text{Gr}(\Lambda/\mathfrak{p}\Lambda, k)$ is the Grassmannian of k -planes in the $A/\mathfrak{p}$ -vector space $\Lambda/\mathfrak{p}\Lambda$.

Proof. We actually have the term-by-term identification

$$1_i^*(1_i)_* = \sum_{x \in H_i \otimes_{A/\mathfrak{p}} (\mathfrak{p}^{-1}A^d/A^d)} t_x^*.$$

Indeed, localizing the lattices at $\mathfrak{p}$, we may choose a basis $e_1, \dots, e_n$ of Λ in which Λ_i has basis $\pi^{-1}e_1, \dots, \pi^{-1}e_k, e_{k+1}, \dots, e_n$. The isogeny 1_i then has kernel given by the first k factors of $E[\mathfrak{p}]$. Intrinsically, under the full Drinfeld trivialization, its kernel cycle is the sum of the sections indexed by

$$H_i \otimes_{A/\mathfrak{p}} (\mathfrak{p}^{-1}A^d/A^d) \subset \Lambda \otimes_A (\mathfrak{p}^{-1}A^d/A^d),$$

with multiplicities supplied by the Drinfeld basis.

To see the term-by-term equality of correspondences, the left-hand side is represented by the subscheme W of $T_E(\Lambda) \times_S T_E(\Lambda)$ cut out by $\{(x, y) \mid 1_i x = 1_i y\}$. Under the difference map $\delta: T_E(\Lambda) \times_S T_E(\Lambda) \rightarrow T_E(\Lambda)$, $(x, y) \mapsto x - y$, this is the preimage of $\ker(1_i)$. Thus

$$[W] = \delta^*[\ker(1_i)] = \sum_{x \in H_i \otimes_{A/\mathfrak{p}} (\mathfrak{p}^{-1}A^d/A^d)} t_x^*$$

as cycles, where we have pulled back our Drinfeld basis identity by the flat map δ . Summing over i gives the proposition, since each k -plane occurs exactly once. $\square$

If E/S does not satisfy the hypothesis of the preceding proposition, then pass to an fppf cover $S(\mathfrak{p}) \rightarrow S$ and a pullback Drinfeld module $E(\mathfrak{p})$ which does; we write π for the projection maps, including for the n -fold Drinfeld module $T_E(\Lambda; \mathfrak{p})$, and in particular

$$\pi \times_{\pi} \pi: T_E(\Lambda; \mathfrak{p}) \times_{S(\mathfrak{p})} T_E(\Lambda; \mathfrak{p}) \rightarrow T_E(\Lambda) \times_S T_E(\Lambda)$$

for the corresponding covering of correspondence spaces over S . Then it follows:

Corollary 4.1.4. *We have the equality of correspondences on $T_E(\Lambda; \mathfrak{p}) \times_{S(\mathfrak{p})} T_E(\Lambda; \mathfrak{p})$*

$$(\pi \times_{\pi} \pi)^* \left(\sum_{i \in I_k(\mathfrak{p})} 1_i^*(1_i)_* \right) = \sum_{H \in \text{Gr}(\Lambda/\mathfrak{p}\Lambda, k)} \sum_{x \in H \otimes_{A/\mathfrak{p}} (\mathfrak{p}^{-1}A^d/A^d)} t_x^*.$$

4.2 Hecke intertwining

We now prove the Hecke intertwining properties of our partial modular symbols Θ_T for T a universal n -fold Drinfeld A -module. We have the fppf-local description Proposition 4.1.3 of correspondences yielding the cohomological Hecke action at a good prime $\mathfrak{p}$ as sums over translations by trivialized $\mathfrak{p}$ -torsion sections; we will give an analogous description of the geometric Hecke actions for the same primes, which will then reduce the intertwining for unramified Hecke operators to understanding aspects of subspace combinatorics in some $A/\mathfrak{p}$ -vector spaces.

4.2.1 fppf-local capitulation of the geometric Hecke action

As with the cohomological operators, it will be very convenient for us to compute the action of $T_k^{(d)}(\mathfrak{p})$ after pullback to a *full level* $\mathfrak{p}$ cover. One may hope that under such a pullback, as in the cohomological setting, the Hecke correspondences on the universal families will decompose into a sum of translation operators - but this is impossible, since the correspondences are not fiber-preserving. However, we will see that the geometric Hecke action on our motivic cohomology classes *does* agree with the action of a *different*, fiber-wise, correspondence, which does have such a decomposition.

Set, as before, $\Phi = \Phi_k(\mathfrak{p})$, $\Psi = \Psi_k(\mathfrak{p})$. Then the action of the geometric Hecke operator $\Phi_* \circ \Psi^*$ on our modular symbol's values can be “straightened” as follows:

Proposition 4.2.1. *Let $\mathcal{E} \rightarrow Y$ be a universal Drinfeld module over a Drinfeld modular curve of level prime to $\mathfrak{p}$; i.e., such that $Y = Y_{\mathcal{L}}^{(d)}$ for $\mathcal{L}$ supported away from $\mathfrak{p}$. Write $\mathcal{E}_0(\mathfrak{p})$ and $\mathcal{E}_1(\mathfrak{p})$ (and similarly for Y) for the corresponding fppf covers with added level structure to $\mathcal{L}$ corresponding to a distinguished $(A/\mathfrak{p})^k$ -subgroup K , respectively a trivialized such subgroup. (This amounts to replacing $\mathcal{L}$ by $\mathcal{L} \oplus \bar{\mathfrak{p}}$ in the latter case, and the quotient of the resulting modular variety by diamond operators in the former.) Then we have*

$$\Psi^* \vartheta = t_{\mathfrak{p}} \circ \Phi^* \vartheta \in H^1(\mathcal{E}_0(\mathfrak{p}) - \{0\}, \mathbb{Z}'(1))$$

where here $t_{\mathfrak{p}}$ is the unique correspondence on $\mathcal{E}_0(\mathfrak{p})$ whose pullback to $\mathcal{E}_1(\mathfrak{p})$ is

$$\sum_{x \in K} t_x^*$$

where K is the canonical subgroup at level $Y_1(\mathfrak{p})$. Equivalently, this is the pushforward along the projection $\mathcal{E}_0(\mathfrak{p}) \times_Y K \rightarrow \mathcal{E}_0(\mathfrak{p})$ of the graph of the addition map $\mathcal{E}_0(\mathfrak{p}) \times_Y K \rightarrow \mathcal{E}_0(\mathfrak{p})$.

Proof. Both sides have residual divisor K , so the equality follows from the fact that both are fixed by all prime-to- $\mathfrak{p}$ trace maps on $\mathcal{E}_0(\mathfrak{p})$ by an identical uniqueness argument to the construction of ϑ in [KoYa1]. $\square$

Analogously to the previous section with full level structure, we use the notation $\mathcal{T}_0(\Lambda; \mathfrak{p})$ and $\mathcal{T}_1(\Lambda; \mathfrak{p})$ (and similar decorations on the $\mathcal{E}$ and Y levels) for the versions with $\hat{\Gamma}_0(\mathfrak{p})$ and $\hat{\Gamma}_1(\mathfrak{p})$ -level structure added to $\mathcal{L}$. We write $t_{\mathfrak{p}}^{\Lambda}$ for the correspondence on $\mathcal{T}_0(\Lambda; \mathfrak{p})$ whose pullback to full level $\mathcal{T}_1(\Lambda; \mathfrak{p})$ is

$$\sum_{x \in \Lambda \otimes_A K} t_x^*.$$

For example, when $\Lambda = A^n$, this is the n -fold fiber product of $t_{\mathfrak{p}}$ over Y .

Corollary 4.2.2. *With the same setup and notation as the previous proposition, for any ideal isogeny $M: \mathcal{E}^n = A^n \otimes_A \mathcal{E} \rightarrow \mathcal{T}(\Lambda) := \mathcal{T}_{\mathcal{E}}(\Lambda)$ whose kernel contains no $\mathfrak{p}$ -torsion, we have*

$$(\Psi^{\Lambda})^* M_*(\vartheta_1 \cup \dots \cup \vartheta_n) = t_{\mathfrak{p}}^{\Lambda} \circ (\Phi^{\Lambda})^* M_*(\vartheta_1 \cup \dots \cup \vartheta_n) \in H^n(\Lambda \otimes_A \mathcal{E} - M_* H, \mathbb{Z}'(n))$$

where H is the union of the hyperplane loci $\{t_i = 0\}$ for $1 \leq i \leq n$.

Proof. For M the identity on $\Lambda = A^n$, this follows formally from the preceding proposition: recall the cup product may be defined as the diagonal pullback of the external product of the elements on n copies of $\mathcal{T}(\Lambda)$, $\mathcal{T}(\Lambda)^{(1)} \times \dots \times \mathcal{T}(\Lambda)^{(n)}$. The operators $\Psi^{A^n}, \Phi^{A^n}, t_{\mathfrak{p}}^{A^n}$ in question then intertwine the diagonal pullback with the factor-wise product of the individual operators $\Psi, t_{\mathfrak{p}}, \Phi$, which satisfy the desired relations by the previous proposition.

For general $M \in \mathrm{GL}_n(A_{\mathfrak{p}}) \cap M_n(A)$ invertible at $\mathfrak{p}$, we wish to commute M_* with $(\Psi^n)^*$ and $t_{\mathfrak{p}}^n \circ (\Phi^n)^* M_*$, respectively. We thus need to check that M_* fits into Cartesian diagrams for these other correspondences: for Φ and Ψ , these are the diagrams

$$\begin{array}{ccc} \mathcal{T}_0(\Lambda; \mathfrak{p}) & \xrightarrow{\Phi^{\Lambda}} & \mathcal{T}(\Lambda) \\ M \downarrow & & \downarrow M \\ \mathcal{T}_0(\Lambda; \mathfrak{p}) & \xrightarrow{\Phi^{\Lambda}} & \mathcal{T}(\Lambda) \end{array} \qquad \begin{array}{ccc} \mathcal{T}_0(\Lambda; \mathfrak{p}) & \xrightarrow{\Psi^{\Lambda}} & \mathcal{T}(\Lambda) \\ M \downarrow & & \downarrow M \\ \mathcal{T}_0(\Lambda; \mathfrak{p}) & \xrightarrow{\Psi^{\Lambda}} & \mathcal{T}(\Lambda) \end{array}$$

which are both essentially tautological on fibers, since M has kernel supported on prime-to- $\mathfrak{p}$ -torsion, on which Φ^{Λ} and Ψ^{Λ} both induce fiberwise isomorphisms.

To check commutativity of M_* with $t_{\mathfrak{p}}^{\Lambda}$, pull back to the auxiliary level $Y_1(\mathfrak{p})$, where we may write the literal equality of Cartier divisors $t_{\mathfrak{p}}^{\Lambda} = \sum_{x \in K \otimes_A \Lambda} t_x^*$ with a sum of pullbacks

along the translations $t_x: E_1(\mathfrak{p})^n \rightarrow E_1(\mathfrak{p})^n$. Each square

$$\begin{array}{ccc} \mathcal{T}_1(\Lambda; \mathfrak{p}) & \xrightarrow{t_x} & \mathcal{T}_1(\Lambda; \mathfrak{p}) \\ \downarrow M & & \downarrow M \\ \mathcal{T}_1(\Lambda; \mathfrak{p}) & \xrightarrow{t_{Mx}} & \mathcal{T}_1(\Lambda; \mathfrak{p}) \end{array} \quad (4.2)$$

commutes, and is obviously Cartesian since the horizontal arrows are isomorphisms. Summing over $x \in K \otimes_A \Lambda$ (using that M is an isomorphism on the $\mathfrak{p}$ -torsion by assumption), we obtain the desired

$$M_* t_{\mathfrak{p}}^{\Lambda} = t_{\mathfrak{p}}^{\Lambda} M_*$$

at this level, which implies the desired identity on $\mathcal{E}_0^{\Lambda}(\mathfrak{p})$ by fppf descent. $\square$

Consider now again the correspondence $\Phi_*^{\Lambda} \circ t_{\mathfrak{p}}^{\Lambda} \circ (\Phi^{\Lambda})^*$ on $\mathcal{T}(\Lambda)$. Since it covers the trivial correspondence on Y , we may consider it as a cycle in $\mathcal{T}(\Lambda) \times_Y \mathcal{T}(\Lambda)$. Denote by $\pi_{\mathfrak{p}} \times_{\pi_{\mathfrak{p}}} \pi_{\mathfrak{p}}$ the fppf covering map $\mathcal{T}(\Lambda; \mathfrak{p}) \times_{Y(\mathfrak{p})} \mathcal{T}(\Lambda; \mathfrak{p}) \rightarrow \mathcal{T}(\Lambda) \times_Y \mathcal{T}(\Lambda)$.

Proposition 4.2.3. *We have an equality of cycles on $\mathcal{T}(\Lambda; \mathfrak{p}) \times_{Y(\mathfrak{p})} \mathcal{T}(\Lambda; \mathfrak{p})$*

$$(\pi_{\mathfrak{p}} \times_{\pi_{\mathfrak{p}}} \pi_{\mathfrak{p}})^*(\Phi_*^{\Lambda} \circ t_{\mathfrak{p}}^{\Lambda} \circ (\Phi^{\Lambda})^*) = \sum_{H \in \text{Gr}((\mathfrak{p}^{-1}A/A)^{d,k})} \sum_{x \in \Lambda \otimes_A H} t_x^*$$

for any $\mathfrak{p}$ not dividing N .

Proof. It suffices to prove this after localizing away from $\mathfrak{p}$, since the identity is of closed cycles flat over the base. (In fact, we could also have used this trick to obtain proofs using only étale instead of fppf descent in the previous results, but they were simple enough that this seemed less necessary.)

This allows us to consider K , and $\mathcal{E}[\mathfrak{p}]$, as étale over Y and $\pi_{\mathfrak{p}}: Y(\mathfrak{p}) \rightarrow Y$ (therefore) a G -torsor for $G = \text{GL}_d(A/\mathfrak{p})$, with $Y_0(\mathfrak{p}) = P \backslash Y(\mathfrak{p})$ for $P \leq G$ a fixed parahoric stabilizer of our stabilized reference subgroup K . The map Φ^{Λ} is a fiberwise isomorphism covering ϕ , and we have the pullback diagram

$$\begin{array}{ccc} \mathcal{T}(\Lambda; \mathfrak{p}) \times_{Y(\mathfrak{p})} (Y(\mathfrak{p}) \times_Y Y_0(\mathfrak{p})) & \xrightarrow{q} & \mathcal{T}_0(\Lambda; \mathfrak{p}) \\ p \downarrow & & \downarrow \Phi^{\Lambda} \\ \mathcal{T}(\Lambda; \mathfrak{p}) & \xrightarrow{\pi_{\mathfrak{p}}} & \mathcal{T}(\Lambda) \end{array}$$

Hence $\pi_{\mathfrak{p}}^*(\Phi^{\Lambda})_* = p_* q^*$, and

$$(\pi_{\mathfrak{p}} \times_{\pi_{\mathfrak{p}}} \pi_{\mathfrak{p}})^*((\Phi^{\Lambda})_* \circ t_{\mathfrak{p}}^{\Lambda} \circ (\Phi^{\Lambda})^*) = (\Phi^{\Lambda})'_* \circ (t_{\mathfrak{p}}^{\Lambda})' \circ ((\Phi^{\Lambda})')^*,$$

the prime denoting base change to $Y(\mathfrak{p})$ along p . Note both p and q are fiberwise isomorphisms; we will (a bit abusively) denote the corresponding maps on the bases by the same letters. We will also write $\rho: Y(\mathfrak{p}) \rightarrow Y_0(\mathfrak{p})$ for the intermediate projection map (as well as its version on the families which is a fiberwise isomorphism), so that $\pi_{\mathfrak{p}} = \phi \circ \rho$. Then the Galois descent map

$$G \times Y(\mathfrak{p}) \xrightarrow{\sim} Y(\mathfrak{p}) \times_Y Y(\mathfrak{p}), \quad (g, a) \mapsto (ga, a),$$

descends to a cover trivialization

$$Y(\mathfrak{p}) \times_Y Y_0(\mathfrak{p}) \cong \coprod_{\gamma \in G/P} Y(\mathfrak{p}),$$

where the γ -indexed copy is $(\gamma a, Pa)$. The restriction of p to this factor is the monodromy action of γ ; q is the projection ρ down to $\Gamma_0(\mathfrak{p})$ -level on all factors. Base changing to $\mathcal{T}(\Lambda)$, we conclude that $(\Phi^\Lambda)'$ acts by the monodromy action of γ on the corresponding γ -factor of

$$\mathcal{T}(\Lambda; \mathfrak{p}) \times_{Y(\mathfrak{p})} (Y(\mathfrak{p}) \times_Y Y_0(\mathfrak{p})) \cong \coprod_{\gamma \in G/P} \mathcal{T}(\Lambda; \mathfrak{p})$$

from which we have the splitting

$$(\Phi^\Lambda)'_* \circ (t_{\mathfrak{p}}^\Lambda)' \circ ((\Phi^\Lambda)')^* = \sum_{\gamma \in G/P} \gamma_* \circ \left(\sum_{x \in \Lambda \otimes_A K} t_x^* \right) \circ \gamma^*.$$

As the sum ranges over G/P , the inner sum exactly is transported to sums over each of the distinct k -planes in the Grassmannian (instead of K , stabilized by P), from which the result follows. $\square$

4.2.2 Some linear algebra

Fix some $\mathcal{X} = (\Lambda, \iota: N^{-1}A/A \rightarrow \Lambda'/\Lambda)$ an object of $\mathbf{Lat}_N(n)$ as before, and fix $\mathfrak{p} \nmid N$. Let $k_{\mathfrak{p}} := A/\mathfrak{p}$ be the residue field at $\mathfrak{p}$, of cardinality $q_{\mathfrak{p}}$, and write $A = \Lambda/\mathfrak{p}\Lambda$ and $B = \mathfrak{p}^{-1}A^d/A^d$ for the respective n -, d -dimensional vector spaces over $k_{\mathfrak{p}}$, temporarily reusing A for the first vector space. Then $W = A \otimes_{k_{\mathfrak{p}}} B$ is the space labeling the $\mathfrak{p}$ -torsion sections of $T_E(\Lambda)$.

Propositions 4.2.3 and 4.1.3 effectively tell us that fppf-locally, both geometric and cohomological Hecke operators can be understood in terms of corresponding linear combinations of torsion sections inside $\mathbb{Z}[W]$. These linear combinations are sums over indicator functions of the form

$$\sum_{x \in H \otimes_{k_{\mathfrak{p}}} B} [x], \quad \sum_{x \in A \otimes_{k_{\mathfrak{p}}} H} [x]$$

for subspaces H of A or B , for the cohomological, respectively geometric, Hecke operators. It is the purpose of this section to understand the relations between these sums of indicator functions, as they will furnish our desired Hecke intertwining properties. All tensor products of these vector spaces are taken over k_p .

The relevant class functions are, for $0 \leq k \leq n$,

$$a_k(\mathfrak{p}) := \sum_{H \in \text{Gr}(A, k)} \sum_{x \in H \otimes B} [x] \in \mathbb{Z}[W],$$

and for $0 \leq k \leq d$,

$$b_k(\mathfrak{p}) := \sum_{H \in \text{Gr}(B, k)} \sum_{x \in A \otimes H} [x] \in \mathbb{Z}[W];$$

we will simply write a_k, b_k, q for $a_k(\mathfrak{p}), b_k(\mathfrak{p}), q_p$ since $\mathfrak{p}$ is always fixed in this section. All of these are fixed vectors under the action of $\text{GL}(A) \times \text{GL}(B)$; thus, their coefficients are constant in the *rank*. In other words, we have an isomorphism

$$\mathbb{Z}[W]^{\text{GL}(A) \times \text{GL}(B)} \cong \bigoplus_{r=0}^{\min\{n, d\}} \mathbb{Z}$$

where the r -labeled copy of $\mathbb{Z}$ corresponds to the indicator function $1_r \in \mathbb{Z}[W]$ which is the sum of $[x]$ over all x which are tensors of rank $0 \leq r \leq \min\{n, d\}$. Writing $a_k[r], b_k[r]$ for the rank- r coefficient of these vectors, one obtains the q_p -binomial coefficient formulas

$$a_k[r] = \binom{n-r}{k-r}_{q_p}, \quad b_k[r] = \binom{d-r}{k-r}_{q_p}$$

since a rank- r tensor is counted in the k -sum once for every k -dimensional subspace containing the corresponding dimension- r subspace in the respective tensor factor. Notice that it is immediately visible that $a_k = b_k$ when $n = d$; this will correspond to the Hecke *equivariance* of our theta modular symbols in this case. More generally, we can describe the relations as follows:

Proposition 4.2.4 (q -Chu–Vandermonde identities). *Let $m = |n - d|$. We have the following two symmetric cases:*

1. *If $d \geq n$, for every $0 \leq \ell \leq d$,*

$$b_\ell = \sum_k q^{(n-k)(\ell-k)} \binom{m}{\ell-k}_q a_k,$$

the sum running over $\max(0, \ell - m) \leq k \leq \min(n, \ell)$.

2. If $n \geq d$, then for every $0 \leq \ell \leq n$,

$$a_\ell = \sum_k q^{(d-k)(\ell-k)} \binom{m}{\ell-k}_q b_k,$$

the sum running over $\max(0, \ell - m) \leq k \leq \min(d, \ell)$.

Note that both cases degenerate to $a_k = b_k$ for all $1 \leq k \leq n = d$ in the overlapping equality case.

Proof. By symmetry, it suffices to prove case 1. Both sides of the claimed identity lie in $\mathbb{Z}[W]^{\text{GL}(A) \times \text{GL}(B)} = \bigoplus_{r \geq 0} \mathbb{Z} \cdot 1_r$, labelled by rank. Using $a_k[r] = \binom{n-r}{k-r}_q$ and $b_\ell[r] = \binom{d-r}{\ell-r}_q$ to check rank r , we are reduced to verifying

$$\binom{d-r}{\ell-r}_q = \sum_k q^{(n-k)(\ell-k)} \binom{m}{\ell-k}_q \binom{n-r}{k-r}_q$$

Split $d - r = (n - r) + m$ and apply the q -Chu–Vandermonde identity

$$\binom{a+b}{c}_q = \sum_j q^{(a-j)(c-j)} \binom{a}{j}_q \binom{b}{c-j}_q$$

with $a = n - r$, $b = m$, $c = \ell - r$:

$$\binom{d-r}{\ell-r}_q = \sum_j q^{(n-r-j)(\ell-r-j)} \binom{n-r}{j}_q \binom{m}{\ell-r-j}_q.$$

The substitution $k = j + r$ sends $n - r - j \mapsto n - k$, sends $\ell - r - j \mapsto \ell - k$, and sends $\binom{n-r}{j}_q \mapsto \binom{n-r}{k-r}_q = a_k[r]$.

The q -Chu–Vandermonde identity itself follows from a standard subspace counting argument: consider a $(d - r)$ -dimensional space over $\mathbb{F}_q$, $V = V_1 \oplus V_2$, with $\dim V_1 = n - r$ and $\dim V_2 = m$; then $\binom{d-r}{\ell-r}_q$ counts the $(\ell - r)$ -dimensional subspaces $H \subseteq V$, which we sort by $i := \dim(H \cap V_1)$. Writing $i = k - r$, the intersection $H \cap V_1$ is one of $\binom{n-r}{k-r}_q$ subspaces of V_1 ; the image $\bar{H}$ of H in $V/V_1 \cong V_2$, of dimension $\ell - k$, is one of $\binom{m}{\ell-k}_q$ subspaces; and the lifts of a fixed $\bar{H}$ to a fixed intersection form a torsor under $\text{Hom}(\bar{H}, V_1/(H \cap V_1))$, of which there are $q^{(n-k)(\ell-k)}$. Multiplying the three factors and summing over k gives the right-hand side. $\square$

4.2.3 Proof of main intertwining theorem

With all the ingredients in place, we now may prove our main theorem:

Theorem 4.2.5. *Let $m = |n - d|$. Then the sections $\mathcal{X} \mapsto \Theta_{\mathcal{T}(\Lambda)}$ and $\mathcal{X} \mapsto s_{\mathcal{X}}^* \Theta_{\mathcal{T}(\Lambda)}$, in $\Gamma(\mathbf{Lat}_N(n)^{\text{iso}}, \mathcal{M})$ and $\Gamma(\mathbf{Lat}_N(n)^{\text{iso}}, s\mathcal{M})$, respectively, satisfy:*

1. *If $d \geq n$, for every $0 \leq \ell \leq d$,*

$$T_{\ell}^{(d)}(\mathfrak{p}) \cdot \Theta_{\mathcal{T}(\Lambda)} = \sum_k q^{(n-k)(\ell-k)} \binom{m}{\ell-k}_q T_k^{(n)}(\mathfrak{p}) \Theta_{\mathcal{T}(\Lambda)}$$

the sum running over $\max(0, \ell - m) \leq k \leq \min(n, \ell)$, for all ideals $\mathfrak{p}$ not dividing N , and similarly for $s^ \Theta_{\mathcal{T}(\Lambda)}$.*

2. *If $n \geq d$, for every $0 \leq \ell \leq n$,*

$$T_{\ell}^{(n)}(\mathfrak{p}) \Theta_{\mathcal{T}(\Lambda)} = \sum_k q^{(d-k)(\ell-k)} \binom{m}{\ell-k}_q T_k^{(d)}(\mathfrak{p}) \cdot \Theta_{\mathcal{T}(\Lambda)}$$

the sum running over $\max(0, \ell - m) \leq k \leq \min(d, \ell)$, for all ideals $\mathfrak{p}$ not dividing N , and similarly for $s^ \Theta_{\mathcal{T}(\Lambda)}$.*

In particular, when $n = d$, both sections intertwine the pairs of individual Hecke operators $T_k^{(n)}(\mathfrak{p})$ and $T_k^{(d)}(\mathfrak{p})$ for each $0 \leq k \leq n = d$, and all primes $\mathfrak{p}$ not dividing the level: i.e., they are equivariant for the action of the anemic Hecke algebra.

Proof. The statement for $s_{\mathcal{X}}^* \Theta_{\mathcal{T}(\Lambda)}$ follows formally from the other one, so it suffices to consider $\mathcal{X} \mapsto \Theta_{\mathcal{T}(\Lambda)}$. Then parts (1) and (2) are proven much the same way: fix an apartment class $\mathcal{C} \in \text{St}(\Lambda^{L_N})$; then

$$\Theta_{\mathcal{T}(\Lambda)}(\mathcal{C}) = M_*(\vartheta_1 \cup \dots \cup \vartheta_n)$$

for some isogeny $M: \mathcal{T}(A^n) \rightarrow \mathcal{T}(\Lambda)$ (which we can view as associated to an isogeny of lattices $A^n \rightarrow \Lambda$). By Remark 4.1.2 and Proposition 2.3.3, it suffices to consider the case where this isogeny is of prime-to- $\mathfrak{p}$ degree, as we may pick a skeletal representative $\mathcal{X}$ of its isomorphism class such that Λ carries a prime-to- $\mathfrak{p}$ isogeny from A^n , and then apartment classes which are $\mathfrak{p}$ -unimodular generate the restricted Steinberg module. We may then compute that

$$\begin{aligned} T_{\ell}^{(d)}(\mathfrak{p}) \cdot \Theta_{\mathcal{T}(\Lambda)}(\mathcal{C}) &= \Phi_*^{\Lambda} \circ (\Psi^{\Lambda})^* \cdot \Theta_{\mathcal{T}(\Lambda)}(\mathcal{C}) \\ &= \Phi_*^{\Lambda} \circ t_{\mathfrak{p}}^{\Lambda} \circ (\Phi^{\Lambda})^* \cdot \Theta_{\mathcal{T}(\Lambda)}(\mathcal{C}) \end{aligned}$$

(where as before we are leaving implicit the $k, \mathfrak{p}$ from the notation of the correspondences) while

$$(T_k^{(n)}(\mathfrak{p}) \Theta_{\mathcal{T}(\Lambda)})(\mathcal{C}) = \left(\sum_{i \in I_k} 1_i^* (1_i)_* \right) \cdot \Theta_{\mathcal{T}(\Lambda)}(\mathcal{C}).$$

Write $\mathcal{A}_k \subset T_{\mathcal{E}}(\Lambda) \times T_{\mathcal{E}}(\Lambda)$ for the cycle $\sum_{i \in I_k} 1_i^*(1_i)_*$ and $\mathcal{B}_k \subset T_{\mathcal{E}}(\Lambda) \times T_{\mathcal{E}}(\Lambda)$ for the cycle $\Phi_*^\Lambda \circ t_{\mathfrak{p}}^\Lambda \circ (\Phi^\Lambda)^*$. Then we claim we have the relations

$$\mathcal{B}_\ell = \sum_k q^{(n-k)(\ell-k)} \binom{m}{\ell-k}_q \mathcal{A}_k$$

or, respectively,

$$\mathcal{A}_\ell = \sum_k q^{(d-k)(\ell-k)} \binom{m}{\ell-k}_q \mathcal{B}_k$$

in the two cases of the theorem. Indeed, Proposition 4.2.4 shows that these relations are true for their pullbacks by $\pi_{\mathfrak{p}}^* \times \pi_{\mathfrak{p}}^*$ using the formulas Corollary 4.1.4 and Proposition 4.2.3; thus, since equality of cycles can be detected fppf-locally, we are done. $\square$

4.3 Other exceptional phenomena in dimension 2

4.3.1 Integrality at bad primes

As in the classical cyclotomic setting, when $n = 2$, restricting the modular symbol $s_{\mathcal{X}}^* \Theta_{T(\mathcal{X})}$ to loops based at a single cusp results in exceptional integrality even at bad primes:

Theorem 4.3.1. *Let $n = 2$; fix an element $\mathcal{X} = (\Lambda', \Lambda, \iota)$ of $\mathbf{Lat}_N(2)$, and fix a cusp c in $\mathbf{P}(\Lambda)$ in the support of elements of $\mathrm{St}(\Lambda^{L_N})$. Then the restriction of $s_{\mathcal{X}}^* \Theta_{T(\mathcal{X})}$ to the submodule $\mathrm{St}(\Lambda, c) \subset \mathrm{St}(\Lambda^{L_N})$, generated by symbols $[\ell_1, \ell_2]$ with both lines in the $\mathrm{Aut}(\mathcal{X})$ -orbit of c , has trivial tame symbol at all primes $\mathfrak{p} \mid N$.*

Proof. Let L_1 and L_2 be the lines (codimension 1 Drinfeld modules) inside $T(\Lambda)$ corresponding to the poles of $\Theta_{T(\mathcal{X})}([\ell_1, \ell_2])$.

Let $\tilde{T}$ be the blowup of $T(\mathcal{X})$ along the zero section, with $b: \tilde{T} \rightarrow T(\mathcal{X})$ the blowdown map with exceptional divisor $D = b^{-1}(\{0\})$, and let $\tilde{\Theta}$ be the pullback of $\Theta_{T(\mathcal{X})}$. We have, formally, $\tilde{s}^* \tilde{\Theta} = s_{\mathcal{X}}^* \Theta_{T(\mathcal{X})}$. We are thus reduced to computing the tame symbols at primes over N of the former.

The support of the tame symbol of $\tilde{\Theta}([\ell_1, \ell_2])$ consists of the strict transform of the two line divisors L_1 and L_2 , and the components of the exceptional divisor D . We claim that the intersections of $\tilde{s}$ with L_1 and L_2 at $\mathfrak{p}$ were transverse originally. Indeed, under the running assumption that neither line contains $s_{\mathcal{X}}$ in the generic fiber, suppose a line is cut out by the kernel of an A -linear functional $\lambda: E \otimes_A \Lambda \rightarrow E'$ where E' is in the ideal isogeny class of the universal Drinfeld module over Y . Then our claim is equivalent to $\lambda(s_{\mathcal{X}})$ intersecting the zero section of E' transversely in the fiber of Y over $\mathfrak{p}$. In order to intersect at all, the torsion section $\lambda(s_{\mathcal{X}})$ must then be a multiple of the tautological section coming from the $\mathfrak{p}$ -power

Drinfeld level structure (for whatever power of p divides N). The completed local ring around any p -point of the base is then one of the universal deformation rings considered in [KoYa3], at which the coordinate function corresponding to this tautological section is a regular parameter by Proposition 2.3.1 in loc. cit.; the claim follows.

Thus, in the blowup, $\tilde{s}$ now *only* may intersect the exceptional divisor. We therefore have $\partial_p \tilde{s}^* \tilde{\Theta}([\ell_1, \ell_2]) = [\tilde{s}_p^* \partial_D \tilde{\Theta}([\ell_1, \ell_2])]^m$ whenever $\tilde{s}$ intersects D over p , where $\tilde{s}_p^*$ is our section restricted to the fiber over p and m is the intersection multiplicity. (Note this pullback is well-defined by the fact that the section avoids the other two polar divisors, from the previous paragraph.) It thus suffices to show that under the theorem hypotheses, $\tilde{s}_p^* \partial_D \tilde{\Theta}([\ell_1, \ell_2])$ vanishes.

To describe the inside tame symbol, we compute using the tautological map $M: T_0 := L_1 \times_S L_2 \rightarrow T(\Lambda)$, which may not be injective if the arrangement of lines fails to be unimodular, but whose kernel can intersect the two lines L_1 and L_2 only at the zero section in the source. Then formally, $M_*\{\theta_1, \theta_2\} = \Theta_{T(\mathcal{X})}$, where θ_i is the Kondo–Yasuda theta function on the Drinfeld module L_i .

Write $P = \ker M$, and write $b_0: \tilde{T}_0 \rightarrow T_0$ for the blowup at P , with exceptional divisor D_0 . Then by blowup functoriality, M lifts to a map $\tilde{M}: \tilde{T}_0 \rightarrow \tilde{T}$, and we have $\tilde{M}_* \partial_{D_0} b_0^* \{\theta_1, \theta_2\} = \partial_D \tilde{\Theta}([\ell_1, \ell_2])$.

The only component of D_0 along which $\{\theta_1, \theta_2\}$ can have a pole is the component over the zero section D'_0 , since P is otherwise disjoint from the two lines in T_0 . This component of the exceptional divisor is the projectivized normal bundle of the zero section in T_0 , so each line in the latter yields a point coming from its corresponding tangent direction; in particular, we may think of the two lines $[L_1]$ and $[L_2]$ as points in D , which are also the same as the intersections of the exceptional divisor with the strict transforms of these lines.

We may thus compute that

$$k(D'_0)^\times \ni \partial_{D'_0} b_0^* \{\theta_1, \theta_2\} = \partial_{D'_0} b_0^* \{\theta_1, -\theta_2/\theta_1\} = -b_0^*(\theta_1/\theta_2)|_{D'_0}.$$

Since we are always inverting 2, the sign can be ignored.

The quotient $b_0^* \theta_1 / b_0^* \theta_2$ may be described intrinsically inside $D'_0 \cong \mathbb{P}(N_{\{0\}/T_0})$ as the quotient $\overline{\theta_1} / \overline{\theta_2}$ of the two sections of the twisting line bundle $\overline{\theta_1}, \overline{\theta_2} \in \Gamma(T_0, I_{\{0\}} / I_{\{0\}}^2) \cong \Gamma(D'_0, \mathcal{O}(1))$ where $I_{\{0\}}$ is the ideal sheaf of the zero section, whose indicated indecomposables are identified with $N_{\{0\}/T_0}^\vee$. (Here, we use that θ_1, θ_2 both vanish at the zero section, but not to quadratic order.)

Now, the restriction of the map M to $D'_0 \rightarrow D$ is an isomorphism (at least generically), so we may write the pushforward as $-\sigma_1/\sigma_2$ where $\sigma_i = M_* \overline{\theta_i}$. We claim that if $\ell_1 = \gamma \ell_2$, then $\sigma_1 = \gamma \sigma_2$, where $\gamma \in \text{Aut}(\mathcal{X})$. From this, the result would follow: we have $\gamma \tilde{s}_p^* = \tilde{s}_p^*$ by

assumption, and γ also fixes the fiber of $\mathcal{O}(1)$ on D since it fixes the entire section $\tilde{s}$ - thus, using a fixed rigidification at this fiber of σ_1 and σ_2 causes the ratio of their evaluations of these sections at $\tilde{s}_p^*$ to cancel.

To prove the claim, the key is the following diagram:

$$\begin{array}{ccc} L_2 & \xrightarrow{\gamma_L} & L_1 \\ j_2 \downarrow & & \downarrow j_1 \\ T(\mathcal{X}) & \xrightarrow{\gamma} & T(\mathcal{X}), \end{array} \quad (4.3)$$

We have $\gamma_L^* \theta_1 = \theta_2 \Rightarrow \gamma_L^* \bar{\theta}_1 = \bar{\theta}_2$. Taking normal bundles in the above diagram gives a commutative diagram

$$\begin{array}{ccc} N_{\{0\}/L_2} & \xrightarrow{d\gamma_L} & N_{\{0\}/L_1} \\ dj_2 \downarrow & & \downarrow dj_1 \\ N_{\{0\}/T(\mathcal{X})} & \xrightarrow{d\gamma} & N_{\{0\}/T(\mathcal{X})}. \end{array} \quad (4.4)$$

After projectivizing, this gives the induced action of γ on the exceptional divisor

$$D = \mathbf{P}(N_{\{0\}/T(\mathcal{X})})$$

together with its natural action on $\mathcal{O}_D(1)$. The two compatibilities then imply $\sigma_1 = \gamma\sigma_2$ as desired. $\square$

4.3.2 The operator $U_1^{(2)}(\mathfrak{p})^*$ when $d = 1$

In addition to the anemic Hecke intertwining results above, we can say something about the ramified cohomological operators when $n = 2$ and $d = 1$ (the direct analogue of the classical cyclotomic setting of the Sharifi conjectures). Let $\mathfrak{p}$ divide N , and work over U_N . Fix a uniformizer idele $\pi_{\mathfrak{p}}$, supported at $\mathfrak{p}$, write π for its $\mathfrak{p}$ -component, and write $q_{\mathfrak{p}} = [A : \mathfrak{p}]$. We would like to consider the cohomological Hecke operator $U_1^{(2)}(\mathfrak{p})^*$ attached to

$$g = \begin{pmatrix} 1 & \\ & \pi_{\mathfrak{p}} \end{pmatrix}.$$

This does not exactly fit into the framework we considered above, as this matrix does not stabilize $\overline{N}$. However, we may define its action in an ad hoc way on $s\mathcal{M}$:

For $\mathcal{X} = (\Lambda, \iota: N^{-1}A/A \rightarrow \Lambda'/\Lambda)$, write

$$\mathcal{X}_{\mathfrak{p}} = \pi_{\mathfrak{p}}^{-1} \cdot \mathcal{X}, \quad Z = \Lambda + \mathfrak{p}^{-1}N\Lambda'.$$

The underlying lattices of $\mathcal{X}_{\mathfrak{p}}$ are $\mathfrak{p}^{-1}\Lambda \subset \mathfrak{p}^{-1}\Lambda'$, and its level structure is $\iota_{\mathfrak{p}} = \pi_{\mathfrak{p}}^{-1} \circ \iota$. We have two inclusion isogenies

$$T_E(\Lambda) \xrightarrow{a} T_E(Z) \xrightarrow{b} T_E(\mathfrak{p}^{-1}\Lambda),$$

with both Z/Λ and $\mathfrak{p}^{-1}\Lambda/Z$ rank-1 over $A/\mathfrak{p}$. Indeed, locally at $\mathfrak{p}$, if N has valuation e and Λ, Λ' have bases (e_1, e_2) and $(e_1, \pi^{-e}e_2)$, respectively, then Z has basis $(e_1, \pi^{-1}e_2)$. In these bases, a and b have matrices $\text{diag}(1, \pi)$ and $\text{diag}(\pi, 1)$, respectively.

Consider all lifts

$$\tilde{\iota}: N^{-1}A/A \rightarrow N^{-1}Z/Z, \quad b \circ \tilde{\iota} = \iota_{\mathfrak{p}},$$

where b on the quotients comes from the inclusion $Z \subset \mathfrak{p}^{-1}\Lambda$. These lifts exist since $N\mathfrak{p}^{-1}\Lambda' \subset Z$, and they form a torsor under $\text{Hom}_A(N^{-1}A/A, \mathfrak{p}^{-1}\Lambda/Z)$. There are thus $q_{\mathfrak{p}}$ -many lifts; write $\mathcal{Z}_{\tilde{\iota}}$ for the object of $\mathbf{Lat}_N(2)$ with the corresponding lift, and $s_{\tilde{\iota}}$ for its distinguished section. Then $b \circ s_{\tilde{\iota}} = s_{\mathcal{X}_{\mathfrak{p}}}$. The identity on F^2 identifies the restricted Steinberg modules for $\mathcal{X}$ and $\mathcal{X}_{\mathfrak{p}}$, and maps this module into the one for each $\mathcal{Z}_{\tilde{\iota}}$.

We may now define the action on a global section σ of $s\mathcal{M}$ by

$$(U_1^{(2)}(\mathfrak{p})^* \sigma)(\mathcal{X})(\mathcal{C}) = \sum_{\tilde{\iota}} \sigma(\mathcal{Z}_{\tilde{\iota}})(\mathcal{C}).$$

This is compatible with isomorphisms of $\mathcal{X}$, so again yields a global section. Note that if we transport g to $\mathcal{X}$ as in (4.1) and write

$$\hat{\Gamma}_{\mathcal{X}} g \hat{\Gamma}_{\mathcal{X}} = \bigsqcup_i \hat{\Gamma}_{\mathcal{X} g_i},$$

the objects $g_i^{-1}\mathcal{X}$ all have underlying lattice Z , with level structures exactly given by the lifts above. For our fixed choice of $\pi_{\mathfrak{p}}$, write $[\mathfrak{p}]$ for the diagonal adelic operator

$$([\mathfrak{p}]\sigma)(\mathcal{X})(\mathcal{C}) = \sigma(\mathcal{X}_{\mathfrak{p}})(\mathcal{C})$$

where the level structure also gets transported by $\pi_{\mathfrak{p}}^{-1}$. Note that if $\mathfrak{p} = (\alpha)$ and one instead uses the principal idele α^{-1} for the scalar translate (and $\text{diag}(1, \alpha)$ for the Hecke operator), multiplication by α would identify the translated object with $\mathcal{X}$, yielding the identity operator.

Then we have the following argument, which is a direct transposition of the argument in the cyclotomic setting of [Xu].

Theorem 4.3.2. *When $d = 1$, for any prime $\mathfrak{p}$ dividing N , the section $\mathcal{X} \mapsto s_{\mathcal{X}}^* \Theta_{\mathcal{T}(\Lambda)}$ of $\Gamma(\mathbf{Lat}_N(2)^{\text{iso}}, s\mathcal{M})$ is annihilated by $U_1^{(2)}(\mathfrak{p})^* - [\mathfrak{p}]$.*

Proof. Since $d = 1$, the q_p sections $s_{\tilde{t}}$ comprise the entire fiber of b over $s_{\mathcal{X}_p}$. Moreover, Theorem 3.1.2 yields $a_*\Theta_{\mathcal{T}(\Lambda)} = \Theta_{\mathcal{T}(Z)}$ and $b_*\Theta_{\mathcal{T}(Z)} = \Theta_{\mathcal{T}(p^{-1}\Lambda)}$. We thus compute:

$$[U_1^{(2)}(\mathfrak{p})^* s^* \Theta](\mathcal{X})(\mathcal{C}) = \sum_{\tilde{t}} s_{\tilde{t}}^* a_* \Theta_{\mathcal{T}(\Lambda)}(\mathcal{C}) \quad (4.5)$$

$$= s_{\mathcal{X}_p}^* b_* a_* \Theta_{\mathcal{T}(\Lambda)}(\mathcal{C}) \quad (4.6)$$

$$= s_{\mathcal{X}_p}^* \Theta_{\mathcal{T}(p^{-1}\Lambda)}(\mathcal{C}) \quad (4.7)$$

$$= [[\mathfrak{p}] s^* \Theta](\mathcal{X})(\mathcal{C}), \quad (4.8)$$

as desired. $\square$

Remark 4.3.3. Formally, one can write down a very similar computation whenever $d = 1$, regardless of what n is, showing that all operators $U_k^{(n)}(\mathfrak{p})$ act by 1. However, this is vacuous when $n > 2$, as the specialized modular symbol is zero: for $n > 2$, we have $H^n(Y, \mathbb{Z}(n)) = 0$ by the norm residue isomorphism theorem and Galois cohomological dimension when Y is an affine curve over a finite field.

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